

Abstract Algebra

Complete Unified Lecture Notes

Groups, Homomorphisms, Permutations, Factor Groups,
Isomorphism Theorems, Commutator Subgroups, and Group
Actions

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Mathematics III

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1 Group Homomorphisms and Isomorphisms

1.1 Definitions and Elementary Examples

Definition 1.1 (Group Homomorphism). Let $(G, \circ)$ and $(G', \circ')$ be two groups. Then a map:

1. $f : G \longrightarrow G'$ is called a **homomorphism** if:

$$f(a \circ b) = f(a) \circ' f(b) \quad \forall a, b \in G.$$

2. $f : G \longrightarrow G'$ is called an **isomorphism** if f is a group homomorphism and a bijective map.

In this case, G is called **isomorphic** to G' , and is denoted as:

$$G \cong G'.$$

Example 1.2 (Zero Homomorphism). Let $(G, \circ)$ and $(G', \circ')$ be two groups. Then the map $f : G \longrightarrow G'$ defined by:

$$f(a) = 0' \quad \forall a \in G$$

(where $0'$ is the identity element of G') is a group homomorphism, known as the **zero homomorphism**.

Verification. Let $a, b \in G$. Then:

$$f(a) = 0', \quad f(b) = 0'.$$

Now:

$$f(a \circ b) = 0' = 0' \circ' 0' = f(a) \circ' f(b).$$

Hence, f is a group homomorphism. □

Example 1.3 (Identity Homomorphism). Let $(G, \circ)$ be a group. The map $I_G : G \longrightarrow G$ defined by:

$$I_G(x) = x \quad \forall x \in G$$

is a group homomorphism, known as the **identity homomorphism**. Moreover, I_G is also an **isomorphism**.

Example 1.4 (Multiplication Map on Integers). Consider the map $f : (\mathbb{Z}, +) \longrightarrow (m\mathbb{Z}, +)$ defined by:

$$f(x) = mx \quad \forall x \in \mathbb{Z}, m \in \mathbb{Z}.$$

Verification. Let $x, y \in \mathbb{Z}$. Then:

$$\begin{aligned} f(x + y) &= m(x + y) \\ &= mx + my \\ &= f(x) + f(y). \end{aligned}$$

$\therefore f$ is a group homomorphism. □

Example 1.5 (A Non-Homomorphism). The map $f : (\mathbb{Z}, +) \longrightarrow (\mathbb{Z}, +)$ defined by:

$$f(x) = 5 \quad \forall x \in \mathbb{Z}.$$

Verification. For any $x, y \in \mathbb{Z}$:

$$f(x) = 5, \quad f(y) = 5.$$

(A) Testing the identity preservation:

$$f(0) = 5 \neq 0. \tag{A}$$

(B) Testing the homomorphism condition:

$$\begin{aligned} f(x+y) &= 5, \\ f(x) + f(y) &= 5 + 5 = 10. \\ \therefore f(x+y) &\neq f(x) + f(y). \end{aligned} \tag{B}$$

From both (A) or (B), we can conclude that f is **not** a group homomorphism. $\square$

Example 1.6 (Logarithmic Homomorphism). Consider the map $f : (\mathbb{R}^+, \cdot) \longrightarrow (\mathbb{R}, +)$ defined by:

$$f(x) = \log_e x \quad \forall x \in \mathbb{R}^+.$$

Verification. Let $x, y \in \mathbb{R}^+$. Then:

$$\begin{aligned} f(x) &= \log_e x, \\ f(y) &= \log_e y. \end{aligned}$$

Now:

$$\begin{aligned} f(x \cdot y) &= \log_e(x \cdot y) \\ &= \log_e x + \log_e y \\ &= f(x) + f(y). \end{aligned}$$

$\therefore f$ is a group homomorphism. $\square$

1.2 Properties of Homomorphisms

Theorem 1.1 (Properties of Homomorphisms). *If $f : (G, \circ) \longrightarrow (G', \circ')$ is a group homomorphism, then:*

1. $f(e) = e'$, where e and e' are the identity elements of G and G' respectively.
2. $f(a^{-1}) = (f(a))^{-1} \quad \forall a \in G$.

Proof. 1. Since $e \circ e = e$, we have:

$$\begin{aligned} &\implies f(e \circ e) = f(e) \\ &\implies f(e) \circ' f(e) = f(e) \circ' e'. \end{aligned}$$

By the left cancellation law in $(G', \circ')$:

$$\implies f(e) = e'.$$

2. Let $a \in G$. Then:

$$a \circ a^{-1} = e = a^{-1} \circ a.$$

Applying f :

$$\begin{aligned} \implies f(a \circ a^{-1}) &= f(e) = f(a^{-1} \circ a) \\ \implies f(a) \circ' f(a^{-1}) &= e' = f(a^{-1}) \circ' f(a) \\ \implies f(a^{-1}) &= (f(a))^{-1}. \end{aligned}$$

□

1.3 Composition of Homomorphisms and Isomorphisms

Theorem 1.2 (Composition of Two Group Homomorphisms). *Composition of two group homomorphisms is a group homomorphism.*

Proof. Let $f : (G, \circ) \longrightarrow (G', \circ')$ and $g : (G', \circ') \longrightarrow (G'', \circ'')$ be two group homomorphisms. Then:

$$g \circ f : (G, \circ) \longrightarrow (G'', \circ'').$$

Let $x, y \in G$, then:

$$\begin{aligned} (g \circ f)(x \circ y) &= g\{f(x \circ y)\} \\ &= g\{f(x) \circ' f(y)\} \quad \{\text{as } f \text{ is a homomorphism}\} \\ &= g(f(x)) \circ'' g(f(y)) \quad \{\text{as } g \text{ is a homomorphism}\} \\ &= (g \circ f)(x) \circ'' (g \circ f)(y). \end{aligned}$$

$\therefore g \circ f$ is a group homomorphism. □

Theorem 1.3 (Composition of Two Isomorphisms). *Composition of two isomorphisms is an isomorphism.*

Proof. Let $f : (G, \circ) \longrightarrow (G', \circ')$ and $g : (G', \circ') \longrightarrow (G'', \circ'')$ be two isomorphisms. Then f and g are homomorphisms and bijective maps.

$$\begin{aligned} \implies g \circ f &\text{ is a homomorphism and bijective map.} \\ \implies g \circ f &\text{ is also a group isomorphism.} \end{aligned}$$

□

Theorem 1.4 (Inverse of an Isomorphism). *If $f : G \longrightarrow G'$ is a group isomorphism, then $f^{-1} : G' \longrightarrow G$ is also a group isomorphism.*

Proof. If f is an isomorphism, then it is a bijective map and group homomorphism. Also, so $f^{-1} : G' \longrightarrow G$ exists and is also bijective.

Let $x', y' \in G'$. Then there must exist unique $x, y \in G$ such that:

$$f(x) = x' \quad \text{and} \quad f(y) = y'.$$

$$[\text{To show: } f^{-1}(x' \circ' y') = f^{-1}(x') \circ f^{-1}(y')]$$

Now:

$$\begin{aligned}
\implies f^{-1}\{x' \circ' y'\} &= f^{-1}\{f(x) \circ' f(y)\} \\
&= f^{-1}\{f(x \circ y)\} \quad \{\text{as } f \text{ is a group homomorphism}\} \\
&= x \circ y \\
&= f^{-1}(x') \circ f^{-1}(y') \quad \{\because f \text{ is bijective}\}.
\end{aligned}$$

So, f^{-1} is a group homomorphism, and hence, an isomorphism. $\square$

Theorem 1.5 (Isomorphism as an Equivalence Relation). *Relation of isomorphism on the set of all groups is an equivalence relation.*

Proof. Let X be the set of all groups, and $G_1, G_2 \in X$. Define a relation $\sim$ on X as:

$$G_1 \sim G_2 \iff G_1 \cong G_2.$$

1. **Reflexive:** Now, let $G \in X$, then the identity map $I_G : G \longrightarrow G$ is an isomorphism. So, $G \cong G$, i.e., $G \sim G$.

$\therefore \sim$ is a reflexive relation on X .

2. **Symmetric:** Let $G_1, G_2 \in X$ such that $G_1 \sim G_2$. So, $\exists$ an isomorphism $f : G_1 \longrightarrow G_2$. Then by the previous result, $f^{-1} : G_2 \longrightarrow G_1$ is an isomorphism. Hence $G_2 \cong G_1$, i.e., $G_2 \sim G_1$.

So, $\sim$ is a symmetric relation on X .

3. **Transitive:** Let $G_1, G_2, G_3 \in X$ such that $G_1 \sim G_2$ and $G_2 \sim G_3$.

$$\implies G_1 \cong G_2 \quad \text{and} \quad G_2 \cong G_3.$$

So, $\exists$ isomorphisms $f : G_1 \longrightarrow G_2$ and $g : G_2 \longrightarrow G_3$. By the earlier result:

$$g \circ f : G_1 \longrightarrow G_3 \text{ is an isomorphism.}$$

Hence $G_1 \cong G_3$, i.e., $G_1 \sim G_3$.

$\therefore \sim$ is a transitive relation on X .

Hence, $\sim$ is an equivalence relation on X . $\square$

1.4 Images and Pre-images of Subgroups

Theorem 1.6 (Images and Pre-images of Subgroups under Homomorphisms). *Let $f : (G, \circ) \longrightarrow (G', \circ')$ be a group homomorphism.*

1. *If $H \leq G$, then $f(H) \leq G'$.*
2. *If $K \leq G'$, then $f^{-1}(K) = \{x \in G \mid f(x) \in K\} \leq G$.*

Proof. 1. **Proof that $f(H) \leq G'$:**

- Since $H \leq G$, $e \in H$. Thus $f(e) = e' \in f(H)$, which implies $f(H) \neq \emptyset$.
- Let $x, y \in f(H)$. Then there exist $a, b \in H$ such that $x = f(a)$ and $y = f(b)$.
- Now evaluate $x \circ' y^{-1}$:

$$\begin{aligned} x \circ' y^{-1} &= f(a) \circ' (f(b))^{-1} \\ &= f(a) \circ' f(b^{-1}) \quad (\text{since } (f(b))^{-1} = f(b^{-1})) \\ &= f(a \circ b^{-1}) \quad (\text{since } f \text{ is a homomorphism}). \end{aligned}$$

- Since $H \leq G$ and $a, b \in H$, we have $a \circ b^{-1} \in H$.
- Therefore, $f(a \circ b^{-1}) \in f(H) \implies x \circ' y^{-1} \in f(H)$.

By the one-step subgroup test, $\boxed{f(H) \leq G'}$.

2. Proof that $f^{-1}(K) \leq G$:

- Since $K \leq G'$, $e' \in K$.
- Since $f(e) = e' \in K$, we have $e \in f^{-1}(K)$, so $f^{-1}(K) \neq \emptyset$.
- Let $x, y \in f^{-1}(K)$. Then by definition, $f(x) \in K$ and $f(y) \in K$.
- Now consider $f(x \circ y^{-1})$:

$$\begin{aligned} f(x \circ y^{-1}) &= f(x) \circ' f(y^{-1}) \\ &= f(x) \circ' (f(y))^{-1}. \end{aligned}$$

- Since $K \leq G'$ and $f(x), f(y) \in K$, we have $f(x) \circ' (f(y))^{-1} \in K$.
- Therefore, $f(x \circ y^{-1}) \in K \implies x \circ y^{-1} \in f^{-1}(K)$.

By the one-step subgroup test, $\boxed{f^{-1}(K) \leq G}$.

□

1.5 Kernel of a Homomorphism

Definition 1.7 (Kernel of a Homomorphism). Let $f : G \longrightarrow G'$ be a group homomorphism. Then the set of all elements of G which are mapped to the identity element of G' is known as the **kernel of f in G** . It is written as $\ker f$:

$$\ker f = \{x \in G \mid f(x) = e'\} = f^{-1}(\{e'\}).$$

Theorem 1.7 (Kernel is a Subgroup of G). *Kernel is a subgroup of G :*

$$\ker f \leq G.$$

Proof. 1. As $f(e) = e' \implies e \in \ker f$. So, $\ker f \neq \emptyset$.

2. Also, by definition, $\ker f \subseteq G$.

3. Let $x, y \in \ker f$.

$$\implies f(x) = e', \quad f(y) = e'.$$

Now evaluate $f(xy^{-1})$:

$$\begin{aligned} f(xy^{-1}) &= f(x)f(y^{-1}) \\ &= f(x)(f(y))^{-1} \\ &= e'(e')^{-1} \\ &= e'e' = e'. \end{aligned}$$

$$\implies xy^{-1} \in \ker f.$$

Therefore, $\boxed{\ker f \leq G}$. □

Theorem 1.8 (A Homomorphism is One-One iff Kernel is Trivial). *A homomorphism $f : G \rightarrow G'$ is one-one if and only if $\ker f = \{e\}$:*

$$f \text{ is one-one} \iff \ker f = \{e\}.$$

Proof. 1. **Direct Implication ($\implies$): Let f is one-one.**

Let $x \in \ker f$.

$$\begin{aligned} \implies f(x) &= e' \\ \implies f(x) &= f(e) \quad (\text{since } f(e) = e') \\ \implies x &= e \quad [\text{as } f \text{ is one-one}]. \end{aligned}$$

So, $\boxed{\ker f = \{e\}}$.

2. **Converse ($\impliedby$): Let $\ker f = \{e\}$.**

To show: f is one-one. Let $f(x) = f(y)$.

$$\begin{aligned} \implies f(x)(f(y))^{-1} &= e' \\ \implies f(xy^{-1}) &= e' \\ \implies xy^{-1} &\in \ker f = \{e\} \\ \implies xy^{-1} &= e \\ \implies x &= y. \end{aligned}$$

Hence, $\boxed{f \text{ is one-one}}$. □

1.6 Examples and Practice Problems on Homomorphisms

Problem 1.8. Let $f : (\mathbb{Z}, +) \rightarrow (\mathbb{Z}, +)$ be defined as $f(x) = 2x$. Is f a homomorphism? Monomorphism? Epimorphism? Isomorphism?

Solution. 1. **Homomorphism:** Let $x, y \in \mathbb{Z}$.

$$f(x + y) = 2(x + y) = 2x + 2y = f(x) + f(y).$$

Thus, f is a homomorphism.

2. Injectivity (Monomorphism):

$$\ker f = \{x \in \mathbb{Z} \mid f(x) = 0\} = \{x \in \mathbb{Z} \mid 2x = 0\} = \{0\}.$$

Since $\ker f = \{0\}$, f is one-to-one (monomorphism).

3. **Surjectivity (Epimorphism):** Let $y = 3 \in \mathbb{Z}$. If $f(x) = 3$, then $2x = 3 \implies x = 3/2 \notin \mathbb{Z}$. Hence, odd integers have no pre-images in $\mathbb{Z}$. Thus, f is **not onto**.

4. **Conclusion:** f is a monomorphism, but **not an isomorphism**. However, $f(\mathbb{Z}) = 2\mathbb{Z}$, so $\mathbb{Z} \cong 2\mathbb{Z}$. □

Problem 1.9. Let $f : (\mathbb{R}, +) \longrightarrow (\mathbb{R}^*, \cdot)$ be defined as $f(x) = 2^x$. Is f an isomorphism?

Solution. 1. **Homomorphism:**

$$f(x + y) = 2^{x+y} = 2^x \cdot 2^y = f(x) \cdot f(y).$$

2. One-to-one:

$$\ker f = \{x \in \mathbb{R} \mid f(x) = 1\} = \{x \in \mathbb{R} \mid 2^x = 1\} = \{0\}.$$

Thus f is 1-1.

3. **Onto:** For any $y \in \mathbb{R}^*$ with $y < 0$ (e.g., $y = -2$), $2^x = -2$ has no real solution. Thus, f is not onto $\mathbb{R}^*$.

4. **Remark:** If the codomain is restricted to $\mathbb{R}^+ = (0, \infty)$, then $f : (\mathbb{R}, +) \rightarrow (\mathbb{R}^+, \cdot)$ is an isomorphism (with inverse $f^{-1}(y) = \log_2 y$). □

Problem 1.10. Let $f : (\mathbb{C}^*, \cdot) \longrightarrow (\mathbb{R}^*, \cdot)$ be defined as $f(z) = |z|$. Show that f is a homomorphism and find $\ker f$.

Solution. 1. For $z_1, z_2 \in \mathbb{C}^*$:

$$f(z_1 \cdot z_2) = |z_1 \cdot z_2| = |z_1| \cdot |z_2| = f(z_1) \cdot f(z_2).$$

Thus f is a group homomorphism.

2. The identity of $(\mathbb{R}^*, \cdot)$ is 1.

$$\ker f = \{z \in \mathbb{C}^* \mid f(z) = 1\} = \{z \in \mathbb{C}^* \mid |z| = 1\} = S^1.$$

Thus, $\ker f$ is the circle group $S^1 = \{e^{i\theta} \mid \theta \in [0, 2\pi)\}$. □

Problem 1.11. Let $f : GL(n, \mathbb{R}) \longrightarrow (\mathbb{R}^*, \cdot)$ be defined as $f(A) = \det(A)$. Show that f is an epimorphism and find $\ker f$.

Solution. 1. For $A, B \in GL(n, \mathbb{R})$:

$$f(AB) = \det(AB) = \det(A) \det(B) = f(A)f(B).$$

Thus f is a group homomorphism.

2. For any $c \in \mathbb{R}^*$, consider the diagonal matrix $D = \text{diag}(c, 1, \dots, 1)$. Then $\det(D) = c \cdot 1 \cdots 1 = c$, so $f(D) = c$. Thus f is surjective.
3. The identity of $\mathbb{R}^*$ is 1.

$$\ker f = \{A \in GL(n, \mathbb{R}) \mid \det(A) = 1\} = SL(n, \mathbb{R}).$$

Thus, the kernel of the determinant map is the Special Linear Group $SL(n, \mathbb{R})$. □

Problem 1.12 (Homomorphism from the Additive Integers to Non-Zero Rationals). Let $f : (\mathbb{Z}, +) \longrightarrow (\mathbb{Q}^*, \cdot)$ be a group homomorphism such that $f(1) = 3$. Determine:

- (a) $f(0)$
- (b) $f(5)$
- (c) $f(-5)$
- (d) A general formula for $f(n)$ for all $n \in \mathbb{Z}$.

Solution. (a) The identity element of the domain $(\mathbb{Z}, +)$ is $e = 0$, and the identity element of the codomain $(\mathbb{Q}^*, \cdot)$ is $e' = 1$. By the fundamental property of group homomorphisms:

$$f(0) = 1.$$

- (b) For any positive integer $n \in \mathbb{Z}^+$, $n = \underbrace{1 + 1 + \cdots + 1}_{n \text{ times}}$. Since f preserves the group operations (mapping addition in $\mathbb{Z}$ to multiplication in $\mathbb{Q}^*$):

$$f(n) = f(1 + 1 + \cdots + 1) = \underbrace{f(1) \cdot f(1) \cdots f(1)}_{n \text{ times}} = [f(1)]^n.$$

Substituting $n = 5$ and $f(1) = 3$:

$$f(5) = [f(1)]^5 = 3^5 = 243.$$

- (c) By the property of homomorphisms on inverses, $f(-a) = [f(a)]^{-1}$ for all $a \in \mathbb{Z}$. Therefore:

$$f(-5) = [f(5)]^{-1} = \frac{1}{f(5)} = \frac{1}{243}.$$

- (d) In general, for any integer $n \in \mathbb{Z}$:

$$f(n) = 3^n.$$

This is well-defined since $3^n \in \mathbb{Q}^*$ for all $n \in \mathbb{Z}$ (since $3^n \neq 0$). □

2 Subgroups, Generators, and Cyclic Groups

2.1 Subgroup Generated by a Subset

Definition 2.1 (Subgroup Generated by a Subset). Let G be a group and let $S \subseteq G$ be any subset of G . The **subgroup generated by S** , denoted by $\langle S \rangle$ or $[S]$, is defined as the smallest subgroup of G containing S :

$$[S] = \bigcap_{\substack{H \leq G \\ S \subseteq H}} H.$$

If $S = \emptyset$, we define $[S] = [\emptyset] = \{e\}$.

Theorem 2.1 (Explicit Structure of the Generated Subgroup $\langle S \rangle$). *Let G be a group and let $S \subseteq G$ be a non-empty subset. Let T be the set of all finite products of elements of S and their inverses:*

$$T = \{a_1^{n_1} a_2^{n_2} \dots a_k^{n_k} \mid k \in \mathbb{N}, a_i \in S, n_i \in \mathbb{Z}\}.$$

Then:

$$\boxed{[S] = T}.$$

Proof. We divide the proof into two main steps:

1. Step 1: Prove that $T \leq G$ containing S .

- For any $s \in S$, $s = s^1 \in T \implies S \subseteq T$.
- In particular, $T \neq \emptyset$.
- Let $x, y \in T$. Then:

$$\begin{aligned} x &= a_1^{n_1} a_2^{n_2} \dots a_r^{n_r} \quad (a_i \in S, n_i \in \mathbb{Z}), \\ y &= b_1^{m_1} b_2^{m_2} \dots b_s^{m_s} \quad (b_j \in S, m_j \in \mathbb{Z}). \end{aligned}$$

- Then:

$$x \circ y = a_1^{n_1} a_2^{n_2} \dots a_r^{n_r} b_1^{m_1} b_2^{m_2} \dots b_s^{m_s} \in T.$$

- Also:

$$\begin{aligned} x^{-1} &= (a_1^{n_1} a_2^{n_2} \dots a_r^{n_r})^{-1} \\ &= (a_r^{n_r})^{-1} (a_{r-1}^{n_{r-1}})^{-1} \dots (a_1^{n_1})^{-1} \\ &= a_r^{-n_r} a_{r-1}^{-n_{r-1}} \dots a_1^{-n_1} \in T. \end{aligned}$$

- Hence, $T \leq G$.

2. Step 2: Prove that $[S] = T$.

- Since T is a subgroup of G containing S , and $[S]$ is the intersection of all subgroups containing S , we have:

$$[S] \subseteq T.$$

- Conversely, let H be any subgroup of G containing S .
- Since $S \subseteq H$ and H is closed under multiplication and inverses, any finite product $a_1^{n_1} a_2^{n_2} \dots a_k^{n_k}$ (with $a_i \in S \subseteq H$) must belong to H .
- Thus, $T \subseteq H$ for every subgroup H containing S .
- Taking the intersection over all such subgroups:

$$T \subseteq \bigcap_{\substack{H \leq G \\ S \subseteq H}} H = [S].$$

Combining both inclusions, we obtain $[S] = T$.

□

2.2 Homomorphic Images of Cyclic Subgroups

Theorem 2.2 (Homomorphic Images of Cyclic Groups are Cyclic). *Let $f : G \rightarrow G'$ be a group homomorphism. If $G = \langle a \rangle$ is cyclic, then $f(G) = \langle f(a) \rangle$ is cyclic.*

Proof. Let $y \in f(G)$. Then $y = f(x)$ for some $x \in G$. Since $G = \langle a \rangle$, there exists $k \in \mathbb{Z}$ such that $x = a^k$.

$$\begin{aligned} y &= f(a^k) \\ &= (f(a))^k \quad (\text{since } f(a^k) = (f(a))^k) \\ &\in \langle f(a) \rangle. \end{aligned}$$

Thus $f(G) \subseteq \langle f(a) \rangle$.

Conversely, let $z \in \langle f(a) \rangle$. Then $z = (f(a))^m$ for some $m \in \mathbb{Z}$.

$$z = (f(a))^m = f(a^m).$$

Since $a^m \in G$, $f(a^m) \in f(G) \implies z \in f(G)$. Thus $\langle f(a) \rangle \subseteq f(G)$. Therefore, $f(G) = \langle f(a) \rangle$. □

2.3 Classification of Cyclic Groups

Theorem 2.3 (Infinite Cyclic Groups). *Every infinite cyclic group is isomorphic to the additive group of integers $(\mathbb{Z}, +)$.*

Proof. Let $G = \langle a \rangle$ be an infinite cyclic group (so $|a| = \infty$). Define $f : (\mathbb{Z}, +) \rightarrow (G, \cdot)$ by:

$$f(k) = a^k \quad \forall k \in \mathbb{Z}.$$

1. **Homomorphism:** Let $k_1, k_2 \in \mathbb{Z}$.

$$\begin{aligned} f(k_1 + k_2) &= a^{k_1 + k_2} \\ &= a^{k_1} \cdot a^{k_2} \\ &= f(k_1) \cdot f(k_2). \end{aligned}$$

Thus, f is a homomorphism.

2. **One-to-one (Injective):**

$$\begin{aligned}\ker f &= \{k \in \mathbb{Z} \mid f(k) = e\} \\ &= \{k \in \mathbb{Z} \mid a^k = e\}.\end{aligned}$$

Since $|a| = \infty$, $a^k = e$ if and only if $k = 0$. Thus $\ker f = \{0\}$, which proves that f is 1-1.

3. **Onto (Surjective):** Let $y \in G$. Since $G = \langle a \rangle$, $y = a^k$ for some $k \in \mathbb{Z}$. Then $f(k) = a^k = y$. Thus f is onto.

Therefore, f is an isomorphism, and $\boxed{G \cong (\mathbb{Z}, +)}$. □

Theorem 2.4 (Finite Cyclic Groups). *Every finite cyclic group of order n is isomorphic to $(\mathbb{Z}_n, +_n)$.*

Proof. Let $G = \langle a \rangle$ with $|G| = |a| = n$. The elements of G are $\{e, a, a^2, \dots, a^{n-1}\}$. The elements of $\mathbb{Z}_n$ are $\{\bar{0}, \bar{1}, \bar{2}, \dots, \overline{n-1}\}$. Define $f : (\mathbb{Z}_n, +_n) \rightarrow (G, \cdot)$ by:

$$f(\bar{r}) = a^r \quad \text{for } 0 \leq r < n.$$

1. **Well-defined and One-to-one:**

$$\begin{aligned}\bar{r}_1 = \bar{r}_2 &\iff r_1 \equiv r_2 \pmod{n} \\ &\iff r_1 - r_2 = qn \quad (q \in \mathbb{Z}) \\ &\iff a^{r_1-r_2} = a^{qn} = (a^n)^q = e^q = e \\ &\iff a^{r_1} \cdot a^{-r_2} = e \\ &\iff a^{r_1} = a^{r_2} \\ &\iff f(\bar{r}_1) = f(\bar{r}_2).\end{aligned}$$

Reading left-to-right establishes well-definedness; right-to-left establishes injectivity.

2. **Homomorphism:** Let $\bar{r}_1, \bar{r}_2 \in \mathbb{Z}_n$. By the division algorithm, $r_1 + r_2 = qn + r$ where $0 \leq r < n$, so $\bar{r}_1 +_n \bar{r}_2 = \bar{r}$.

$$\begin{aligned}f(\bar{r}_1 +_n \bar{r}_2) &= f(\bar{r}) = a^r \\ &= a^{r_1+r_2-qn} \\ &= a^{r_1} \cdot a^{r_2} \cdot (a^n)^{-q} \\ &= a^{r_1} \cdot a^{r_2} \cdot e^{-q} \\ &= a^{r_1} \cdot a^{r_2} \\ &= f(\bar{r}_1) \cdot f(\bar{r}_2).\end{aligned}$$

Thus f is a homomorphism.

3. **Onto:** Any element of G is of the form a^r with $0 \leq r < n$, and $f(\bar{r}) = a^r$. Thus f is onto.

Therefore, $\boxed{G \cong (\mathbb{Z}_n, +_n)}$. □

3 The Center of a Group

3.1 Definition and Subgroup Proof

Definition 3.1 (Centre of a Group). Let G be a group. The collection of all elements of G which commute with every element of G is called the **centre of G** , denoted by $Z(G)$:

$$Z(G) = \{z \in G \mid zx = xz \quad \forall x \in G\}.$$

Theorem 3.1 (The Center $Z(G)$ is a Characteristic Subgroup of G). *For any group G , $Z(G) \leq G$.*

Proof. 1. **Non-empty:** Since $ex = xe = x$ for all $x \in G$, $e \in Z(G)$. Thus $Z(G) \neq \emptyset$.

2. **Subgroup Criterion:** Let $a, b \in Z(G)$. Then for all $x \in G$:

$$\begin{aligned} ax &= xa, \\ bx &= xb. \end{aligned}$$

From $bx = xb$, multiplying on the left and right by b^{-1} :

$$\begin{aligned} b^{-1}(bx)b^{-1} &= b^{-1}(xb)b^{-1} \\ \implies (b^{-1}b)xb^{-1} &= b^{-1}x(bb^{-1}) \\ \implies exb^{-1} &= b^{-1}xe \\ \implies xb^{-1} &= b^{-1}x. \end{aligned}$$

Thus $b^{-1} \in Z(G)$.

Now consider $(ab^{-1})x$:

$$\begin{aligned} (ab^{-1})x &= a(b^{-1}x) \\ &= a(xb^{-1}) \quad (\text{since } b^{-1} \in Z(G)) \\ &= (ax)b^{-1} \\ &= (xa)b^{-1} \quad (\text{since } a \in Z(G)) \\ &= x(ab^{-1}). \end{aligned}$$

Therefore, $ab^{-1} \in Z(G)$.

By the one-step subgroup test, $\boxed{Z(G) \leq G}$. □

3.2 Properties and Examples of Centers

Theorem 3.2 (Properties of $Z(G)$). 1. $Z(G)$ is always an abelian subgroup of G .

2. G is abelian if and only if $Z(G) = G$.

3. $|Z(G)| \geq 1$ for all groups G .

Example 3.2 (Centers of Common Groups). 1. **Dihedral Group D_4 :** The group of symmetries of a square has order 8:

$$D_4 = \{R_0, R_{90}, R_{180}, R_{270}, H, V, D, D'\}.$$

The Cayley table of D_4 shows that R_0 and R_{180} commute with all 8 elements, whereas rotations by 90° and reflections do not commute with all elements (e.g., $R_{90}H = D \neq D' = HR_{90}$). Thus:

$$Z(D_4) = \{R_0, R_{180}\}.$$

2. **Symmetric Group S_3 :** $S_3 = \{I, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$. Testing elements:

$$(1\ 2)(1\ 2\ 3) = (2\ 3) \neq (1\ 3) = (1\ 2\ 3)(1\ 2).$$

No non-identity element commutes with all elements. Thus:

$$Z(S_3) = \{I\}.$$

3. **Quaternion Group Q_8 :** $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$. $ij = k = -ji$, but 1 and -1 commute with everything. Thus:

$$Z(Q_8) = \{1, -1\}.$$

4 Permutation Groups and the Symmetric Group S_n

4.1 Permutations and Cycle Notation

Definition 4.1 (Permutation and Symmetric Group). Let A be a non-empty set. A bijection $\sigma : A \rightarrow A$ is called a **permutation** on A . The set of all permutations on A forms a group under function composition, called the **symmetric group on A** , denoted by S_A or $\text{Sym}(A)$. When $|A| = n$, we denote it by S_n , with:

$$|S_n| = n!$$

Definition 4.2 (Cycles and Transpositions). • A **k -cycle** $\sigma = (a_1\ a_2\ \dots\ a_k)$ is a permutation mapping $a_1 \mapsto a_2 \mapsto \dots \mapsto a_k \mapsto a_1$ and fixing all other elements. The integer k is the **length** of the cycle.

- A 2-cycle $(a\ b)$ is called a **transposition**.
- Two cycles $\sigma = (a_1\ \dots\ a_k)$ and $\tau = (b_1\ \dots\ b_m)$ are **disjoint** if $\{a_1, \dots, a_k\} \cap \{b_1, \dots, b_m\} = \emptyset$.

Theorem 4.1 (Commutativity of Disjoint Cycles). *Disjoint cycles commute: if $\sigma, \tau \in S_n$ are disjoint cycles, then $\sigma\tau = \tau\sigma$.*

Theorem 4.2 (Disjoint Cycle Decomposition). *Every permutation $\sigma \in S_n$ can be uniquely expressed (up to the order of factors) as a product of disjoint cycles.*

Theorem 4.3 (Order of a Permutation). *The order of a permutation $\sigma \in S_n$ expressed in disjoint cycle form is the least common multiple of the lengths of its disjoint cycles:*

$$|\sigma| = \text{lcm}(\text{length of } c_1, \text{length of } c_2, \dots, \text{length of } c_k).$$

Example 4.3 (Order of a Permutation in S_6 via LCM of Cycle Lengths). In S_6 , let $\sigma = (1\ 2\ 3)(4\ 5)$. The cycle lengths are 3 and 2.

$$|\sigma| = \text{lcm}(3, 2) = 6.$$

4.2 Parity, Alternating Group A_n , and Klein 4-Group V_4

Theorem 4.4 (Factorization of Cycles into Transpositions). *Every cycle of length k can be factored into a product of $k - 1$ transpositions:*

$$(a_1 \ a_2 \ \dots \ a_k) = (a_1 \ a_k)(a_1 \ a_{k-1}) \dots (a_1 \ a_3)(a_1 \ a_2).$$

Definition 4.4 (Even and Odd Permutations). A permutation is **even** if it can be represented as a product of an even number of transpositions; it is **odd** if it can be represented as a product of an odd number of transpositions. No permutation is both even and odd.

Definition 4.5 (Alternating Group A_n). The set of all even permutations in S_n forms a subgroup called the **Alternating Group of degree n** , denoted by A_n :

$$|A_n| = \frac{n!}{2} \quad \text{for } n \geq 2.$$

Definition 4.6 (Klein Four-Group V_4). In S_4 , the subset:

$$V_4 = \{I, (1 \ 2)(3 \ 4), (1 \ 3)(2 \ 4), (1 \ 4)(2 \ 3)\}$$

is an abelian subgroup of order 4 in which every non-identity element has order 2. Moreover, $V_4 \leq A_4 \leq S_4$.

4.3 Conjugacy in S_n and Cycle Counting

Definition 4.7 (Conjugate Permutations). Two permutations $p, q \in S_n$ are **conjugate** if there exists $r \in S_n$ such that $q = rpr^{-1}$.

Theorem 4.5 (Conjugacy Criterion via Cycle Structure in S_n). *Two permutations in S_n are conjugate if and only if they have the identical cycle structure (same cycle lengths).*

Example 4.8 (Finding the Conjugating Permutation r). Let $p = (1 \ 2 \ 3)(4 \ 5 \ 6)(7 \ 8)$ and $q = (2 \ 3 \ 4)(5 \ 1 \ 8)(6 \ 7)$ in S_8 . Writing p and q with aligned cycle components:

$$\begin{aligned} p &= (1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8), \\ q &= (2 \ 3 \ 4 \ 5 \ 1 \ 8 \ 6 \ 7). \end{aligned}$$

Define $r \in S_8$ by sending the top row to the bottom row:

$$r = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 3 & 4 & 5 & 1 & 8 & 6 & 7 \end{pmatrix}.$$

Then $q = rpr^{-1}$.

Theorem 4.6 (Formula for Number of r -Cycles in S_n). *The total number of distinct cycles of length r in S_n is:*

$$\boxed{\binom{n}{r}(r-1)! = \frac{{}^nP_r}{r} = \frac{n!}{r(n-r)!}}.$$

5 Cayley's Representation Theorem

Theorem 5.1 (Cayley's Theorem). *Every group G is isomorphic to a subgroup of the symmetric group S_G . If $|G| = n$, then G is isomorphic to a subgroup of S_n .*

Proof. For each $g \in G$, define the left-multiplication mapping $\lambda_g : G \rightarrow G$ by:

$$\lambda_g(x) = gx \quad \forall x \in G.$$

1. **Step 1:** λ_g is a permutation on G ($\lambda_g \in S_G$).

- *Injective:* Let $\lambda_g(x) = \lambda_g(y) \implies gx = gy \implies x = y$ (by left cancellation in G).
- *Surjective:* Let $y \in G$. Then $\lambda_g(g^{-1}y) = g(g^{-1}y) = (gg^{-1})y = ey = y$.

Thus, λ_g is a bijection of G onto itself, so $\lambda_g \in S_G$.

2. **Step 2:** Define $\phi : G \rightarrow S_G$ by $\phi(g) = \lambda_g$.

- ϕ is a homomorphism: For any $g, h \in G$ and $x \in G$:

$$\begin{aligned} \lambda_{gh}(x) &= (gh)x = g(hx) \\ &= \lambda_g(hx) = \lambda_g(\lambda_h(x)) \\ &= (\lambda_g \circ \lambda_h)(x). \end{aligned}$$

Since this holds for all $x \in G$, $\lambda_{gh} = \lambda_g \circ \lambda_h$. Thus:

$$\phi(gh) = \phi(g) \circ \phi(h).$$

- ϕ is one-to-one (injective):

$$\begin{aligned} \ker \phi &= \{g \in G \mid \phi(g) = I_G\} \\ &= \{g \in G \mid \lambda_g = I_G\} \\ &= \{g \in G \mid \lambda_g(x) = x \quad \forall x \in G\} \\ &= \{g \in G \mid gx = x \quad \forall x \in G\}. \end{aligned}$$

Evaluating at $x = e$:

$$ge = e \implies g = e.$$

Thus $\ker \phi = \{e\}$, which proves that ϕ is 1-1.

Therefore, $\phi : G \rightarrow \phi(G)$ is an isomorphism of G onto $\phi(G) \leq S_G$. Hence, $G \cong \phi(G) \leq S_G$. □

6 Cosets and Lagrange's Theorem

6.1 Left and Right Cosets

Definition 6.1 (Cosets). Let $H \leq G$ and $a \in G$.

- The **left coset** of H in G determined by a is $aH = \{ah \mid h \in H\}$.

- The **right coset** of H in G determined by a is $Ha = \{ha \mid h \in H\}$.

Theorem 6.1 (Properties of Cosets). *Let $H \leq G$ and $a, b \in G$.*

1. $a \in aH$ (since $a = ae \in aH$).
2. $aH = H \iff a \in H$.
3. $aH = bH \iff a^{-1}b \in H \iff b \in aH$.
4. $Ha = Hb \iff ab^{-1} \in H \iff a \in Hb$.
5. Either $aH = bH$ or $aH \cap bH = \emptyset$.
6. $|aH| = |H| = |Ha|$ for all $a \in G$.
7. The collection of all left (or right) cosets forms a partition of G .

Proof that Left Cosets Partition G . 1. **Union is G :** For each $x \in G$, $x = xe \in xH \subseteq \bigcup_{g \in G} gH$. Thus $G = \bigcup_{g \in G} gH$.

2. **Pairwise Disjoint:** Suppose $aH \cap bH \neq \emptyset$. Let $x \in aH \cap bH$. Then $x = ah_1 = bh_2$ for some $h_1, h_2 \in H$.

$$a = bh_2h_1^{-1}.$$

Since $h_1, h_2 \in H$, $h_2h_1^{-1} = h_3 \in H$. Thus $a = bh_3 \in bH \implies aH = bH$.

Thus, distinct left cosets are mutually disjoint and cover G . $\square$

Proposition 6.2 (Coset Subgroup Criterion). *Let $H \leq G$ and $a \in G$. Then the left coset aH is a subgroup of G if and only if $a \in H$ (in which case $aH = H$).*

Proof. ($\implies$) Suppose $aH \leq G$. Every subgroup must contain the identity element e . Thus $e \in aH$. Also, since $H \leq G$, $e \in H = eH$. Hence $aH \cap eH \neq \emptyset$. Since any two left cosets are either identical or disjoint, $aH \cap eH \neq \emptyset \implies aH = eH = H$. Since $a \in aH = H$, we conclude that $a \in H$.

($\impliedby$) Conversely, if $a \in H$, then by coset absorption, $aH = H$. Since $H \leq G$ by assumption, $aH = H$ is a subgroup of G . $\square$

Example 6.2 (Distinction Between Left and Right Cosets in S_3). Consider the symmetric group on 3 elements $S_3 = \{I, (12), (13), (23), (123), (132)\}$ and the non-normal subgroup of order 2:

$$H = \{I, (12)\}.$$

We compute the left and right cosets of H with respect to the 3-cycle (123) :

- **Left Coset:**

$$(123)H = \{(123) \circ I, (123) \circ (12)\} = \{(123), (13)\}.$$

- **Right Coset:**

$$H(123) = \{I \circ (123), (12) \circ (123)\} = \{(123), (23)\}.$$

Notice that:

$$(1\ 2\ 3)H = \{(1\ 2\ 3), (1\ 3)\} \neq \{(1\ 2\ 3), (2\ 3)\} = H(1\ 2\ 3).$$

This explicitly demonstrates that for a non-normal subgroup, a left coset and a right coset determined by the same element need not be equal.

Theorem 6.3 (Bijection Between Left and Right Cosets). *Let G be a group (finite or infinite) and let $H \leq G$. Let $(G/H)^l$ denote the set of all left cosets of H in G , and let $(G/H)^r$ denote the set of all right cosets of H in G . Then there exists a bijection:*

$$F : (G/H)^l \longrightarrow (G/H)^r \quad \text{defined by} \quad F(aH) = Ha^{-1}.$$

Consequently, the number of distinct left cosets equals the number of distinct right cosets:

$$|(G/H)^l| = |(G/H)^r| = [G : H].$$

Proof. We verify the required conditions:

1. **Well-defined and Injective:** Let $a, b \in G$. We establish the following chain of equivalent statements:

$$\begin{aligned} aH = bH &\iff a^{-1}b \in H \\ &\iff (a^{-1}b)^{-1} \in H \quad (\text{since } H \leq G \text{ is closed under inverses}) \\ &\iff b^{-1}(a^{-1})^{-1} \in H \\ &\iff b^{-1}a \in H \\ &\iff Hb^{-1} = Ha^{-1} \\ &\iff Ha^{-1} = Hb^{-1} \\ &\iff F(aH) = F(bH). \end{aligned}$$

The forward implication ($\implies$) proves that F is well-defined (independent of the choice of representative). The reverse implication ($\impliedby$) proves that F is injective (one-to-one).

2. **Surjective (Onto):** Let $Ha \in (G/H)^r$ be an arbitrary right coset. Since $a \in G$, its inverse $a^{-1} \in G$. Thus $a^{-1}H \in (G/H)^l$ is a well-defined left coset. Evaluating F on this left coset:

$$F(a^{-1}H) = H(a^{-1})^{-1} = Ha.$$

Hence every right coset has a pre-image under F , proving that F is surjective.

Therefore, F is a bijection, establishing that the collection of left cosets and the collection of right cosets have the same cardinality for any subgroup $H \leq G$. $\square$

6.2 Lagrange's Theorem and the Index of a Subgroup

Theorem 6.4 (Lagrange's Theorem). *Let G be a finite group and let H be a subgroup of G . Then the order of H divides the order of G :*

$$|H| \mid |G|.$$

Moreover, the number of distinct left (or right) cosets of H in G , called the **index** of H in G and denoted by $[G : H]$, is:

$$[G : H] = \frac{|G|}{|H|}.$$

Proof. Let $|G| = n$ and $|H| = m$. Since the distinct left cosets $g_1H, g_2H, \dots, g_kH$ partition G (where $k = [G : H]$):

$$G = g_1H \cup g_2H \cup \dots \cup g_kH \quad (\text{disjoint union}).$$

Taking cardinality on both sides:

$$|G| = |g_1H| + |g_2H| + \dots + |g_kH|.$$

Since $|g_iH| = |H| = m$ for every $i \in \{1, 2, \dots, k\}$:

$$|G| = k \cdot |H| = [G : H] \cdot |H|.$$

Dividing by $|H|$ gives $[G : H] = \frac{|G|}{|H|}$, and $|H|$ divides $|G|$. □

Corollary 6.5 (Fundamental Consequences). *1. **Element Order Divides Group***

Order: For every $a \in G$, $|a| \mid |G|$. *Proof:* Consider the cyclic subgroup $H = \langle a \rangle$. Then $|H| = |a|$. By Lagrange's theorem, $|H| \mid |G| \implies |a| \mid |G|$.

*2. **Power Equals Identity:** For every $a \in G$, $a^{|G|} = e$. Proof:* Let $|G| = n$ and $|a| = m$. Since $m \mid n$, $n = mq$ for some $q \in \mathbb{Z}$. Then $a^n = a^{mq} = (a^m)^q = e^q = e$.

*3. **Prime Order Groups are Cyclic:** Every group of prime order p is cyclic and simple (has no non-trivial proper subgroups). Proof:* Let $|G| = p$ (prime). Let $a \in G$ with $a \neq e$. Consider $H = \langle a \rangle \leq G$. Since $a \neq e$, $|H| > 1$. By Lagrange's theorem, $|H| \mid p \implies |H| = p = |G| \implies G = \langle a \rangle$.

6.3 Number-Theoretic Theorems

Theorem 6.6 (Euler's Theorem). Let $a, n \in \mathbb{Z}^+$ with $\gcd(a, n) = 1$. Then:

$$a^{\phi(n)} \equiv 1 \pmod{n},$$

where $\phi(n)$ is Euler's totient function.

Proof. Consider the group of units $U(n) = (\mathbb{Z}_n^*, \cdot)$ under multiplication modulo n . $|U(n)| = \phi(n)$. For $\gcd(a, n) = 1$, the residue class $\bar{a} \in U(n)$. By Lagrange's corollary, $\bar{a}^{|U(n)|} = \bar{1} \implies \bar{a}^{\phi(n)} = \bar{1} \implies a^{\phi(n)} \equiv 1 \pmod{n}$. □

Theorem 6.7 (Fermat's Little Theorem). If p is prime and $a \in \mathbb{Z}$ with $p \nmid a$, then:

$$a^{p-1} \equiv 1 \pmod{p} \implies a^p \equiv a \pmod{p} \quad \forall a \in \mathbb{Z}.$$

Theorem 6.8 (Wilson's Theorem). An integer $p > 1$ is prime if and only if:

$$(p-1)! \equiv -1 \pmod{p}.$$

Theorem 6.9 (Converse of Wilson's Theorem). Let $n > 1$ be a positive integer. If $(n-1)! \equiv -1 \pmod{n}$, then n is a prime number.

Proof. We proceed by contradiction. Assume that n is not prime (i.e., n is composite). Then n can be factored as:

$$n = r \cdot s \quad \text{where } 1 < r < n \text{ and } 1 < s < n.$$

Since $1 < r < n$, we have $2 \leq r \leq n-1$. Therefore, r is one of the factors appearing in the product $(n-1)! = 1 \cdot 2 \cdots (n-1)$, which implies:

$$r \mid (n-1)!.$$

On the other hand, the given congruence $(n-1)! \equiv -1 \pmod{n}$ means that:

$$n \mid ((n-1)! + 1).$$

Since $r \mid n$, by transitivity of divisibility:

$$r \mid ((n-1)! + 1).$$

Now, since r divides both $(n-1)! + 1$ and $(n-1)!$, it must divide their difference:

$$r \mid \left(((n-1)! + 1) - (n-1)! \right) \implies r \mid 1.$$

The only positive divisor of 1 is 1, so $r = 1$. This contradicts the strict inequality $r > 1$. Hence our initial assumption that n is composite is false. Therefore, n must be a prime number. $\square$

7 Normal Subgroups and Conjugacy

7.1 Definition and Equivalent Characterizations

Definition 7.1 (Normal Subgroup). A subgroup $H \leq G$ is called a **normal subgroup** of G , denoted by:

$$\boxed{H \trianglelefteq G},$$

if for all $g \in G$ and all $h \in H$, $ghg^{-1} \in H$ (or equivalently, $gHg^{-1} = H$ for all $g \in G$).

Theorem 7.1 (Equivalent Conditions for Normality). *Let $H \leq G$. The following statements are equivalent:*

1. $H \trianglelefteq G$.
2. $gHg^{-1} \subseteq H$ for all $g \in G$.
3. $gH = Hg$ for all $g \in G$ (left cosets equal right cosets).
4. $(aH)(bH) = (ab)H$ defines a well-defined binary operation on cosets.

Theorem 7.2 (Normality in Abelian Groups). *If G is an abelian group, then every subgroup H of G is a normal subgroup ($H \trianglelefteq G$).*

Proof. Let $H \leq G$ and assume G is abelian. For any $a \in G$ and any $h \in H$, since G is commutative:

$$ah = ha.$$

Therefore:

$$aH = \{ah \mid h \in H\} = \{ha \mid h \in H\} = Ha \quad \forall a \in G.$$

Since every left coset coincides with the corresponding right coset, $H \trianglelefteq G$. $\square$

Remark (Failure of the Converse: Hamiltonian Groups and Q_8). The converse of the above theorem is **false**: if every subgroup of a group G is normal, G is not necessarily abelian. A standard counterexample is the **quaternion group** $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$:

- Q_8 is non-abelian, since $ij = k \neq -k = ji$.
- The subgroups of Q_8 are:

$$\{1\}, \quad \{\pm 1\}, \quad \langle i \rangle = \{\pm 1, \pm i\}, \quad \langle j \rangle = \{\pm 1, \pm j\}, \quad \langle k \rangle = \{\pm 1, \pm k\}, \quad Q_8.$$

The trivial subgroup and Q_8 are always normal. The center $Z(Q_8) = \{\pm 1\}$ is always normal. The cyclic subgroups $\langle i \rangle, \langle j \rangle, \langle k \rangle$ each have index $[Q_8 : \langle x \rangle] = \frac{8}{4} = 2$, and subgroups of index 2 are always normal.

Thus, **every subgroup of Q_8 is normal**, yet Q_8 is non-abelian. A non-abelian group in which every subgroup is normal is called a **Hamiltonian group**.

Theorem 7.3 (Conjugate Subgroups). *Let G be a group and $H \leq G$. For any fixed element $x \in G$, the conjugate set:*

$$xHx^{-1} = \{xhx^{-1} \mid h \in H\}$$

*is a subgroup of G , called the **conjugate subgroup** of H with respect to x .*

Proof. We apply the one-step subgroup criterion:

1. **Non-emptiness:** Since $H \leq G$, the identity element $e \in H$. Then:

$$e = xex^{-1} \in xHx^{-1}.$$

Thus xHx^{-1} is non-empty and contains the identity of G .

2. **Closure under ab^{-1} :** Let $a, b \in xHx^{-1}$. Then:

$$a = xh_1x^{-1} \quad \text{and} \quad b = xh_2x^{-1} \quad \text{for some } h_1, h_2 \in H.$$

Computing the inverse of b :

$$b^{-1} = (xh_2x^{-1})^{-1} = (x^{-1})^{-1}h_2^{-1}x^{-1} = xh_2^{-1}x^{-1}.$$

Now multiplying a and b^{-1} :

$$\begin{aligned} ab^{-1} &= (xh_1x^{-1})(xh_2^{-1}x^{-1}) \\ &= xh_1(x^{-1}x)h_2^{-1}x^{-1} \\ &= x(h_1h_2^{-1})x^{-1}. \end{aligned}$$

Since H is a subgroup of G and $h_1, h_2 \in H$, we have $h_1h_2^{-1} \in H$. Therefore, $ab^{-1} = x(h_1h_2^{-1})x^{-1} \in xHx^{-1}$.

By the one-step subgroup test, $xHx^{-1} \leq G$. □

Problem 7.2 (Subgroups Containing All Squares are Normal). Let G be a group and let $H \leq G$. If $a^2 \in H$ for every $a \in G$, prove that $H \trianglelefteq G$.

Proof. To establish that $H \trianglelefteq G$, we must show that $aha^{-1} \in H$ for all $a \in G$ and all $h \in H$. Let $a \in G$ and $h \in H$.

1. Since $a, h \in G$, their product $ah \in G$. By the given hypothesis that every element squared lies in H :

$$(ah)^2 \in H \implies ahah \in H.$$

2. Since $h \in H$ and H is a subgroup, $h^{-1} \in H$. Multiplying the element $ahah \in H$ by $h^{-1} \in H$ on the right:

$$(ahah)h^{-1} = aha(hh^{-1}) = aha(e) = aha \in H.$$

3. Since $a \in G$, its inverse $a^{-1} \in G$. Applying the square hypothesis to a^{-1} :

$$(a^{-1})^2 \in H \implies a^{-2} \in H.$$

4. Now multiply $aha \in H$ by $a^{-2} \in H$ on the right:

$$(aha)a^{-2} = ah(aa^{-2}) = aha^{-1} \in H.$$

Thus, $aha^{-1} \in H$ for all $a \in G$ and all $h \in H$. By the conjugate characterization of normality, $H \trianglelefteq G$. □

Problem 7.3 (Isomorphism Between Conjugate Subgroups and Unique Subgroup Normality). Let G be a group and let $H \leq G$.

- (a) If H is finite, prove that $|xHx^{-1}| = |H|$ for all $x \in G$.
- (b) Prove that if H is the unique subgroup of G of a given finite order m , then $H \trianglelefteq G$.

Proof. (a) Define the mapping $\psi : H \longrightarrow xHx^{-1}$ by:

$$\psi(h) = xhx^{-1} \quad \forall h \in H.$$

- **Injective:** Let $h_1, h_2 \in H$ with $\psi(h_1) = \psi(h_2)$.

$$xh_1x^{-1} = xh_2x^{-1}.$$

Multiplying by x^{-1} on the left and x on the right:

$$x^{-1}(xh_1x^{-1})x = x^{-1}(xh_2x^{-1})x \implies h_1 = h_2.$$

Thus ψ is one-to-one.

- **Surjective:** By definition, every element of xHx^{-1} is of the form $xhx^{-1} = \psi(h)$ for some $h \in H$. Thus ψ is onto.

Since ψ is a bijection, $|xHx^{-1}| = |H|$.

- (b) Suppose H is the unique subgroup of G of order m , so $|H| = m$. For any $x \in G$, by part (a), the conjugate set xHx^{-1} is a subgroup of G with:

$$|xHx^{-1}| = |H| = m.$$

Since H is the *only* subgroup of G having order m , xHx^{-1} must be identical to H :

$$xHx^{-1} = H \quad \forall x \in G.$$

Therefore, $H \trianglelefteq G$. □

Theorem 7.4 (Preservation of Normality under Epimorphisms). *Let $f : G \rightarrow G'$ be a group epimorphism (surjective homomorphism).*

1. *If $H \trianglelefteq G$, then $f(H) \trianglelefteq G'$.*
2. *If $H' \trianglelefteq G'$, then $f^{-1}(H') \trianglelefteq G$.*

Proof. 1. Let $t \in f(H)$ and $y \in G'$. Since $t \in f(H)$, there exists $h \in H$ such that $t = f(h)$. Since f is surjective and $y \in G'$, there exists $x \in G$ such that $y = f(x)$. Since $H \trianglelefteq G$, $xhx^{-1} \in H$. Evaluating f on this conjugate element:

$$\begin{aligned} f(xhx^{-1}) &= f(x)f(h)f(x^{-1}) \\ &= f(x)f(h)(f(x))^{-1} \\ &= yty^{-1}. \end{aligned}$$

Since $xhx^{-1} \in H$, its image $yty^{-1} = f(xhx^{-1}) \in f(H)$. Since this holds for all $y \in G'$ and $t \in f(H)$, we conclude $f(H) \trianglelefteq G'$.

2. Let $s \in f^{-1}(H')$ and $z \in G$. By definition of pre-image, $f(s) \in H'$. Also $f(z) \in G'$. Since $H' \trianglelefteq G'$, the conjugate of $f(s)$ by $f(z)$ remains in H' :

$$f(z)f(s)(f(z))^{-1} \in H'.$$

Since f is a homomorphism:

$$f(z)f(s)(f(z))^{-1} = f(zsz^{-1}) \in H'.$$

By definition of the pre-image:

$$zsz^{-1} \in f^{-1}(H').$$

Since this holds for all $z \in G$ and $s \in f^{-1}(H')$, we conclude $f^{-1}(H') \trianglelefteq G$. □

7.2 Subgroups of Index 2 are Normal

Theorem 7.5 (Subgroups of Index 2 are Normal). *If $H \leq G$ such that $[G : H] = 2$, then $H \trianglelefteq G$.*

Proof. Since $[G : H] = 2$, there are exactly two left cosets and two right cosets of H in G . Let $g \in G$.

- If $g \in H$, then $gH = H = Hg$.
- If $g \notin H$, the left cosets are H and gH . Since cosets partition G :

$$G = H \cup gH \implies gH = G \setminus H.$$

Similarly, the right cosets are H and Hg , so:

$$G = H \cup Hg \implies Hg = G \setminus H.$$

Therefore, $gH = G \setminus H = Hg$.

Thus $gH = Hg$ for all $g \in G$. Hence, $\boxed{H \trianglelefteq G}$. □

Example 7.4 (Normality of the Alternating Group $A_n \trianglelefteq S_n$). In the symmetric group S_n , since $|A_n| = \frac{n!}{2}$ and $|S_n| = n!$:

$$[S_n : A_n] = \frac{n!}{n!/2} = 2 \implies \boxed{A_n \trianglelefteq S_n \quad \forall n \geq 2}.$$

7.3 Intersections and Non-Transitivity of Normality

Theorem 7.6 (Intersections and Direct Images of Normal Subgroups). 1. If $\{H_i\}_{i \in I}$ is a family of normal subgroups of G , then $\bigcap_{i \in I} H_i \trianglelefteq G$.

2. If $f : G \rightarrow G'$ is a homomorphism, then $\ker f \trianglelefteq G$.

3. If $H \leq G$ and $K \trianglelefteq G$, then $H \cap K \trianglelefteq H$.

Proof of (3). Since $H \leq G$ and $K \trianglelefteq G$, $H \cap K \leq H$. Let $x \in H \cap K$ and $h \in H$.

- $x \in H, h \in H \implies h x h^{-1} \in H$.
- $x \in K, h \in G, \text{ and } K \trianglelefteq G \implies h x h^{-1} \in K$.

Thus $h x h^{-1} \in H \cap K$. Therefore, $H \cap K \trianglelefteq H$. □

Note (Normality is Not Transitive). $K \trianglelefteq H$ and $H \trianglelefteq G$ does **not** imply $K \trianglelefteq G$. *Counterexample in S_4 :* Let $V_4 = \{I, (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\} \trianglelefteq S_4$. Let $H = \{I, (1\ 2)(3\ 4)\} \leq V_4$. Since $[V_4 : H] = 2$, $H \trianglelefteq V_4$. However, H is not normal in S_4 , because:

$$(1\ 3)(1\ 2)(3\ 4)(1\ 3)^{-1} = (2\ 3)(1\ 4) \notin H.$$

7.4 Normalizer and Centralizer

Definition 7.5 (Centralizer and Normalizer of a Subgroup). Let $H \leq G$ and $a \in G$.

- **Normalizer of H in G :** $N_G(H) = \{g \in G \mid gHg^{-1} = H\}$.
- **Centralizer of H in G :** $C_G(H) = \{g \in G \mid gh = hg \ \forall h \in H\}$.
- **Centralizer of an element a :** $C_G(a) = \{g \in G \mid ga = ag\}$.

Theorem 7.7 (Properties of Centralizers and Normalizers). 1. $C_G(H) \leq N_G(H) \leq G$.

2. $H \trianglelefteq N_G(H)$, and $H \trianglelefteq G \iff N_G(H) = G$.

3. The size of the conjugacy class of a is $|\text{cl}(a)| = [G : C_G(a)]$.

8 The Correspondence Theorem for Group Homomorphisms

8.1 Completion of the Correspondence Theorem for Group Homomorphisms

Let $(G, \circ)$ and $(G', \circ')$ be two groups, and let $f : G \rightarrow G'$ be a group epimorphism (i.e., a surjective group homomorphism).

Let $\mathcal{S}(G)$ denote the set of all subgroups of G containing $\ker f$:

$$\mathcal{S}(G) = \{H \leq G \mid \ker f \subseteq H\},$$

and let $\mathcal{S}(G')$ denote the set of all subgroups of G' :

$$\mathcal{S}(G') = \{K \leq G'\}.$$

Define the mapping $\phi : \mathcal{S}(G) \rightarrow \mathcal{S}(G')$ by:

$$\phi(H) = f(H) \quad \text{for every } H \in \mathcal{S}(G).$$

Theorem 8.1 (Surjectivity of the Correspondence Map). *The mapping ϕ is surjective (onto).*

Proof. Let $K \in \mathcal{S}(G')$, which means that K is a subgroup of G' ($K \leq G'$). We must show that there exists some $H \in \mathcal{S}(G)$ such that $\phi(H) = K$.

Consider the complete pre-image of K under f :

$$H = f^{-1}(K) = \{x \in G \mid f(x) \in K\}.$$

We verify the required properties step-by-step:

1. **$f^{-1}(K)$ is a subgroup of G :** Since $e' \in K$ (where e' is the identity of G'), $f(e) = e' \in K$, so $e \in f^{-1}(K) \neq \emptyset$. Let $x, y \in f^{-1}(K)$. Then $f(x) \in K$ and $f(y) \in K$. Since $K \leq G'$,

$$f(xy^{-1}) = f(x)(f(y))^{-1} \in K \implies xy^{-1} \in f^{-1}(K).$$

Thus, $f^{-1}(K) \leq G$.

2. **$\ker f$ is contained in $f^{-1}(K)$:** Let $x \in \ker f$. Then $f(x) = e'$. Since $K \leq G'$, the identity element e' belongs to K . Thus:

$$f(x) = e' \in K \implies x \in f^{-1}(K).$$

Therefore:

$$\ker f \subseteq f^{-1}(K).$$

Consequently, $f^{-1}(K) \in \mathcal{S}(G)$.

3. **Image under ϕ :** Applying ϕ to $f^{-1}(K)$:

$$\phi(f^{-1}(K)) = f(f^{-1}(K)).$$

Since $f : G \rightarrow G'$ is given to be surjective (onto), for every element $k \in K \subseteq G'$, there exists $g \in G$ such that $f(g) = k$. By definition, $g \in f^{-1}(K)$, so $f(f^{-1}(K)) = K$. Thus:

$$\phi(f^{-1}(K)) = K.$$

Hence, ϕ is onto. Since ϕ was previously established to be injective, ϕ is a bijection. $\square$

9 Normal Subgroup Intersections and Conjugacy in Permutation Groups

9.1 Intersection of a Subgroup and a Normal Subgroup

Theorem 9.1 (Intersection of a Subgroup and a Normal Subgroup). *Let G be a group, and let $H, K \leq G$. If K is a normal subgroup of G ($K \trianglelefteq G$), then:*

$$H \cap K \trianglelefteq H.$$

Proof. We proceed in two explicit steps:

1. **$H \cap K$ is a subgroup of H :** Since $H \leq G$ and $K \leq G$, the intersection of two subgroups is always a subgroup of G . Furthermore, $H \cap K \subseteq H$, so clearly:

$$H \cap K \leq H.$$

2. **$H \cap K$ is invariant under conjugation by elements of H :** Let $t \in H \cap K$ and let $h \in H$. By definition of set intersection:

$$t \in H \quad \text{and} \quad t \in K.$$

- Since $t \in H$ and $h \in H$, and H is a subgroup of G , by closure under multiplication and inverses:

$$hth^{-1} \in H.$$

- Since $h \in H \subseteq G$ and $t \in K$, and K is given to be a normal subgroup of G ($K \trianglelefteq G$), every conjugate of an element of K by any element of G remains in K :

$$hth^{-1} \in K \quad (\text{as } K \trianglelefteq G).$$

Combining both deductions:

$$hth^{-1} \in H \quad \text{and} \quad hth^{-1} \in K \implies hth^{-1} \in H \cap K.$$

Since for all $h \in H$ and all $t \in H \cap K$ we have $hth^{-1} \in H \cap K$, it follows by definition of normality that:

$$\therefore H \cap K \trianglelefteq H. \quad \square$$

9.2 Properties and Non-Transitivity of Normality

Note. 1. **Normality is NOT a transitive relation:** If $K \trianglelefteq H$ and $H \trianglelefteq G$, then K need not be a normal subgroup of G :

$$K \trianglelefteq H \text{ and } H \trianglelefteq G \not\Rightarrow K \trianglelefteq G.$$

(A concrete counterexample in S_4 is analyzed in Section 10.2).

2. **Intermediate normality:** If $H \trianglelefteq G$ and K is any intermediate subgroup of G containing H ($H \leq K \leq G$), then:

$$H \trianglelefteq K.$$

Proof. For all $k \in K$, since $K \subseteq G$, we have $k \in G$. Since $H \trianglelefteq G$, $kHk^{-1} = H$. Thus $kHk^{-1} = H$ holds for all $k \in K$, proving $H \trianglelefteq K$. $\square$

9.3 Conjugacy in the Symmetric Group S_n

Definition 9.1 (Conjugate Permutations). Two permutations p and $q \in S_n$ are said to be **conjugate** to each other in S_n if there exists a permutation $r \in S_n$ such that:

$$q = rpr^{-1}.$$

Definition 9.2 (Permutations of the Same Form / Cycle Type). Two permutations $p, q \in S_n$ are said to be of the **same form** (or have the same **cycle structure**) if:

1. Both p and q can be expressed as a product of the same number of disjoint cycles.
2. The corresponding disjoint cycles are of equal length.

Theorem 9.2 (Conjugacy Equivalence of Permutations in S_n). *Two permutations $p, q \in S_n$ are permutations of the same form if and only if they are conjugate in S_n .*

Example 9.3 (Finding the Conjugating Permutation in S_8). Let $p, q \in S_8$ be defined by:

$$\begin{aligned} p &= (1\ 2\ 3)(4\ 5\ 6)(7\ 8), \\ q &= (6\ 7)(2\ 3\ 4)(5\ 1\ 8). \end{aligned}$$

Observe that:

- p consists of two 3-cycles and one 2-cycle.
- q consists of two 3-cycles and one 2-cycle, since disjoint cycles commute:

$$q = (2\ 3\ 4)(5\ 1\ 8)(6\ 7).$$

Thus p and q have the exact same cycle form. By the theorem, they are conjugate. To find $r \in S_8$ such that $q = rpr^{-1}$, align the cycle representations of p and q directly:

$$\begin{aligned} p &= (1\ 2\ 3)(4\ 5\ 6)(7\ 8) \\ q &= (2\ 3\ 4)(5\ 1\ 8)(6\ 7) \end{aligned}$$

We define the permutation r by mapping each element of p to the corresponding element directly below it in q :

$$r = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 3 & 4 & 5 & 1 & 8 & 6 & 7 \end{pmatrix}.$$

Expressing r in disjoint cycle notation:

$$r = (1\ 2\ 3\ 4\ 5)(6\ 8\ 7).$$

Its inverse r^{-1} is obtained by reversing each cycle:

$$r^{-1} = (1\ 5\ 4\ 3\ 2)(6\ 7\ 8) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 5 & 1 & 2 & 3 & 4 & 7 & 8 & 6 \end{pmatrix}.$$

Verification of $rpr^{-1} = q$: Writing in two-row matrix form:

$$\begin{aligned}
rpr^{-1} &= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 3 & 4 & 5 & 1 & 8 & 6 & 7 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 3 & 1 & 5 & 6 & 4 & 8 & 7 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 5 & 1 & 2 & 3 & 4 & 7 & 8 & 6 \end{pmatrix} \\
&= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 3 & 4 & 5 & 1 & 8 & 6 & 7 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 6 & 2 & 3 & 1 & 5 & 8 & 7 & 4 \end{pmatrix} \\
&= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 8 & 3 & 4 & 2 & 1 & 7 & 6 & 5 \end{pmatrix} \\
&= (185)(234)(67) \\
&= (234)(518)(67) = q.
\end{aligned}$$

Hence, $q = rpr^{-1}$ is verified.

10 Subgroups of S_4 , Inverses, and Failure of the Converse of Lagrange's Theorem

10.1 Explicit Structure of S_4 and the Alternating Group A_4

The symmetric group S_4 consists of all $4! = 24$ permutations on $\{1, 2, 3, 4\}$. Written out in full two-row and disjoint cycle forms:

$$S_4 = \left\{ I, \right. \\ (2\ 3\ 4), (2\ 4\ 3), (3\ 4), (2\ 3), (2\ 4), (1\ 2), (1\ 2)(3\ 4), \\ (1\ 2\ 3), (1\ 2\ 3\ 4), (1\ 2\ 4\ 3), (1\ 2\ 4), (1\ 3\ 2), \\ (1\ 3\ 4\ 2), (1\ 3), (1\ 3\ 4), (1\ 3)(2\ 4), (1\ 3\ 2\ 4), \\ \left. (1\ 4\ 3\ 2), (1\ 4), (1\ 4\ 2), (1\ 4\ 3), (1\ 4\ 2\ 3), (1\ 4)(2\ 3) \right\}.$$

The alternating group A_4 is the subgroup of all even permutations in S_4 . Since $|S_4| = 24$, its order is:

$$|A_4| = \frac{|S_4|}{2} = \frac{24}{2} = 12.$$

Explicitly, A_4 contains the identity, the eight 3-cycles, and the three products of two disjoint transpositions:

$$A_4 = \left\{ I, \right. \\ (1\ 2\ 3), (1\ 3\ 2), (1\ 2\ 4), (1\ 4\ 2), (1\ 3\ 4), (1\ 4\ 3), (2\ 3\ 4), (2\ 4\ 3), \\ \left. (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3) \right\}.$$

Since $[S_4 : A_4] = 2$, it is immediate that:

$$A_4 \trianglelefteq S_4.$$

10.2 The Klein Four-Group and Non-Transitivity of Normality

The **Klein four-group** embedded in S_4 is:

$$V_4 = \{I, (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\}.$$

Notice that:

- $V_4 \subseteq A_4$.
- V_4 contains all permutations of cycle structure $(2, 2)$ in S_4 along with the identity.
- Since conjugation preserves cycle structure, for any $g \in S_4$ and any $v \in V_4$, gvg^{-1} must also have cycle structure $(2, 2)$ or be the identity, hence $gvg^{-1} \in V_4$.

Therefore:

$$V_4 \trianglelefteq S_4 \quad \text{and} \quad V_4 \trianglelefteq A_4.$$

Example 10.1 (Counterexample: Normality is NOT Transitive). Consider the subgroup H of V_4 generated by a single double transposition:

$$H = \{I, (1\ 2)(3\ 4)\} \leq V_4.$$

1. The index of H in V_4 is:

$$[V_4 : H] = \frac{|V_4|}{|H|} = \frac{4}{2} = 2.$$

Since every subgroup of index 2 is normal:

$$H \trianglelefteq V_4.$$

2. We already have $V_4 \trianglelefteq S_4$.
3. However, H is **not** a normal subgroup of S_4 ($H \not\trianglelefteq S_4$).

Proof. Take $g = (1\ 3) \in S_4$. Then $g^{-1} = (1\ 3)$. Conjugating $(1\ 2)(3\ 4) \in H$ by g :

$$g(1\ 2)(3\ 4)g^{-1} = (1\ 3)(1\ 2)(3\ 4)(1\ 3) = (3\ 2)(1\ 4) = (1\ 4)(2\ 3).$$

Notice that $(1\ 4)(2\ 3) \notin H$. Thus $gHg^{-1} \not\subseteq H$. □

Therefore, $H \trianglelefteq V_4$ and $V_4 \trianglelefteq S_4$, but $H \not\trianglelefteq S_4$.

Note. In S_n , if $H \trianglelefteq S_n$ and $p \in H$, then all permutations in S_n that have the same cycle form as p must also belong to H (since normal subgroups are unions of complete conjugacy classes).

10.3 Elements of Order 2 in Finite Groups of Even Order

Theorem 10.1 (Cauchy's Lemma for $p = 2$: Elements of Order 2 in Even-Order Groups). *If G is a finite group of even order, then there exists an element $a \in G$ such that $a \neq e$ and $a^2 = e$ (i.e., $|a| = 2$).*

Proof. Let G be a finite group with $|G| = 2k$ for some positive integer k . Every element $x \in G$ has a unique inverse $x^{-1} \in G$.

An element x satisfies $x = x^{-1}$ if and only if $x^2 = e$. The identity element e always satisfies $e = e^{-1}$ (i.e., $e^2 = e$).

Now consider the set of non-identity elements:

$$G \setminus \{e\}.$$

The cardinality of this set is:

$$|G \setminus \{e\}| = |G| - 1 = 2k - 1,$$

which is an **odd** integer.

We can partition $G \setminus \{e\}$ into two disjoint subsets:

1. Elements that are not their own inverses:

$$A = \{x \in G \setminus \{e\} \mid x \neq x^{-1}\}.$$

These elements can be grouped into disjoint pairs $\{x, x^{-1}\}$ of size 2. Consequently, $|A|$ is an **even** integer.

2. Elements that are their own inverses:

$$B = \{x \in G \setminus \{e\} \mid x = x^{-1}\}.$$

Since $(G \setminus \{e\}) = A \cup B$ and $A \cap B = \emptyset$:

$$|B| = |G \setminus \{e\}| - |A| = (\text{odd}) - (\text{even}) = \text{odd}.$$

Since $|B|$ is an odd integer, $|B| \geq 1$. Therefore, there exists at least one element $a \in G \setminus \{e\}$ such that:

$$a = a^{-1} \implies a^2 = e.$$

This element a has order $|a| = 2$. □

10.4 Failure of the Converse of Lagrange's Theorem

Lagrange's Theorem asserts that if G is a finite group and $H \leq G$, then $|H|$ divides $|G|$ ($|H| \mid |G|$).

The **converse of Lagrange's Theorem** would state:

If G is a finite group and $n \mid |G|$, then G must have a subgroup of order n .

The converse is false in general.

Example 10.2 (A_4 Has No Subgroup of Order 6). Consider the alternating group A_4 . We have $|A_4| = 12$. The integer 6 divides 12 ($6 \mid 12$). We prove that A_4 has **no** subgroup of order 6.

Proof. Suppose, for the sake of contradiction, that H is a subgroup of A_4 of order 6 ($|H| = 6$).

1. The index of H in A_4 is:

$$[A_4 : H] = \frac{|A_4|}{|H|} = \frac{12}{6} = 2.$$

Since any subgroup of index 2 is normal:

$$H \trianglelefteq A_4.$$

2. Since $|H| = 6$ is an even integer, by the preceding theorem, there exists an element $p \in H$ such that:

$$p \neq I \quad \text{and} \quad p^2 = I \quad (\text{i.e., } |p| = 2).$$

3. Recall that the elements of order 2 in A_4 are precisely the three double transpositions:

$$(1\ 2)(3\ 4), \quad (1\ 3)(2\ 4), \quad (1\ 4)(2\ 3).$$

Without loss of generality, let $p = (1\ 2)(3\ 4) \in H$.

4. Since $H \trianglelefteq A_4$, H is closed under conjugation by every element of A_4 .

- Conjugating p by $(1\ 2\ 3) \in A_4$:

$$(1\ 4)(2\ 3) = (1\ 2\ 3)(1\ 2)(3\ 4)(1\ 2\ 3)^{-1} \in H.$$

- Conjugating p by $(1\ 2\ 4) \in A_4$:

$$(1\ 3)(2\ 4) = (1\ 2\ 4)(1\ 2)(3\ 4)(1\ 2\ 4)^{-1} \in H.$$

5. Since $H \trianglelefteq A_4$, both $(1\ 4)(2\ 3)$ and $(1\ 3)(2\ 4)$ must also belong to H . Together with $I \in H$ and $p = (1\ 2)(3\ 4) \in H$, we have:

$$V_4 = \{I, (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\} \subseteq H.$$

6. Since V_4 is a subgroup of A_4 contained in H , $V_4 \leq H$. By Lagrange's Theorem, the order of V_4 must divide the order of H :

$$|V_4| \mid |H| \implies 4 \mid 6,$$

which is an outright contradiction, since $4 \nmid 6$.

Therefore, our assumption that A_4 contains a subgroup of order 6 is false. There is no subgroup of A_4 of order 6. $\square$

11 Quotient Groups (Factor Groups) and Canonical Projection

11.1 Coset Multiplication and Well-Definedness

Let G be a group and let $H \leq G$. The set of all left cosets of H in G is denoted by:

$$G/H = \{aH \mid a \in G\}.$$

We define a candidate binary operation $*$: $G/H \times G/H \rightarrow G/H$ by:

$$*(aH, bH) = abH \quad \text{or} \quad aH * bH = abH.$$

Remark (Failure of Coset Multiplication for Non-Normal Subgroups). The operation $*$ **need not be a well-defined mapping** if H is merely a subgroup of G and not normal!

Example 11.1 (Counterexample: Operation Not Well-Defined When $H \not\trianglelefteq G$). Consider the symmetric group $G = S_3$ and the subgroup $H = \{I, (12)\}$. Compute the left cosets:

$$\begin{aligned} (123)H &= \{(123)I, (123)(12)\} = \{(123), (13)\} = (13)H, \\ (23)H &= \{(23)I, (23)(12)\} = \{(23), (132)\}. \end{aligned}$$

Since $(123)H = (13)H$, for $*$ to be well-defined, the product with $(23)H$ must yield the exact same coset:

$$\text{Must have:} \quad (123)H * (23)H = (13)H * (23)H.$$

Let us compute each side explicitly using the formula $xH * yH = xyH$:

- Left side:

$$(123)H * (23)H = (123)(23)H = (12)H = H.$$

- Right side:

$$(13)H * (23)H = (13)(23)H = (132)H = \{(132), (23)\} \neq H.$$

Therefore:

$$(123)H * (23)H \neq (13)H * (23)H.$$

Thus, the operation $*$ depends on the choice of coset representative, and so is **not well-defined**.

Theorem 11.1 (Well-Definedness Criterion and Quotient Group). *Let G be a group and $H \trianglelefteq G$. Then the operation:*

$$*: G/H \times G/H \rightarrow G/H, \quad aH * bH = abH$$

*is a well-defined binary operation on G/H . Furthermore, $(G/H, *)$ is a group, known as the **quotient group** (or **factor group**) of G with respect to H .*

Proof. We divide the proof into two parts:

Part 1: Well-Definedness. Let $(aH, bH) = (a'H, b'H)$ in $G/H \times G/H$. This means:

$$aH = a'H \quad \text{and} \quad bH = b'H.$$

We must show that $abH = a'b'H$.

- $aH = a'H \implies a \in a'H \implies a = a'h_1$ for some $h_1 \in H$.
- $bH = b'H \implies b \in b'H \implies b = b'h_2$ for some $h_2 \in H$.

Now calculate the coset abH :

$$\begin{aligned} abH &= (a'h_1)(b'h_2)H \\ &= a'h_1b'(h_2H) && (\text{as } h_2 \in H \implies h_2H = H) \\ &= a'h_1b'H. \end{aligned}$$

Since $H \trianglelefteq G$, left and right cosets coincide, so $b'H = Hb'$. Thus, $h_1b' \in Hb' = b'H$, which implies that there exists $h_3 \in H$ such that:

$$h_1b' = b'h_3.$$

Substituting this back:

$$\begin{aligned} abH &= a'(h_1b')H \\ &= a'(b'h_3)H \\ &= a'b'(h_3H) && (\text{as } h_3 \in H \implies h_3H = H) \\ &= a'b'H. \end{aligned}$$

Hence:

$$abH = a'b'H \implies aH * bH = a'H * b'H.$$

So $*$ is indeed a well-defined map, i.e., a binary operation on G/H .

Part 2: Group Axioms for $(G/H, *)$.

1. **Associativity:** Let $aH, bH, cH \in G/H$.

$$\begin{aligned} (aH * bH) * cH &= (abH) * cH \\ &= (ab)cH \\ &= a(bc)H && (\text{associativity in } G) \\ &= aH * (bcH) \\ &= aH * (bH * cH). \end{aligned}$$

So $*$ is associative.

2. **Identity Element:** The coset $eH = H$ serves as the identity element in G/H , because for any $aH \in G/H$:

$$aH * eH = (ae)H = aH \quad \text{and} \quad eH * aH = (ea)H = aH.$$

Thus $eH = H$ is the identity element in G/H with respect to $*$.

3. **Inverses:** For any $aH \in G/H$, the coset $a^{-1}H \in G/H$ is its inverse, because:

$$aH * a^{-1}H = (aa^{-1})H = eH = H \quad \text{and} \quad a^{-1}H * aH = (a^{-1}a)H = eH = H.$$

Thus, $(aH)^{-1} = a^{-1}H$.

Hence, $(G/H, *)$ is a group, known as the quotient group of G with respect to H . □

Note. If G is an abelian group, then the quotient group G/H is also abelian.

Proof. For any $aH, bH \in G/H$:

$$aH * bH = abH = baH = bH * aH \quad (\text{since } ab = ba \text{ in } G). \quad \square$$

11.2 The Canonical (Natural) Projection

Theorem 11.2 (The Canonical Projection Homomorphism). *Let G be a group and $H \trianglelefteq G$. Then the mapping:*

$$\nu : G \rightarrow G/H, \quad \nu(x) = xH \quad \forall x \in G$$

is a surjective group homomorphism with $\ker \nu = H$.

Proof. 1. **ν is a group homomorphism:** Let $x, y \in G$. Then:

$$\nu(xy) = (xy)H = xH * yH = \nu(x) * \nu(y).$$

Thus $\nu(xy) = \nu(x)\nu(y)$, so ν is a group homomorphism.

2. **ν is surjective (onto):** Let $t \in G/H$. By definition of G/H , t is a left coset of the form:

$$t = aH = \nu(a) \quad \text{for some } a \in G.$$

So ν is surjective. Hence ν is a surjective group homomorphism.

3. **Computation of $\ker \nu$:** By definition of the kernel of a homomorphism:

$$\begin{aligned} \ker \nu &= \{x \in G \mid \nu(x) = e_{G/H}\} \\ &= \{x \in G \mid \nu(x) = H\} \\ &= \{x \in G \mid xH = H\} \\ &= \{x \in G \mid x \in H\} \\ &= H. \end{aligned}$$

Thus, $\ker \nu = H$. □

Remark (Significance of the Natural Homomorphism). This homomorphism ν is known as the **canonical projection**, **natural homomorphism**, or **quotient homomorphism**.

11.3 The Correspondence Theorem for Quotient Groups

Note. Recall from set theory that if $f : X \rightarrow Y$ is a mapping and $A_1, A_2 \subseteq X$:

1. $f(A_1 \cup A_2) = f(A_1) \cup f(A_2)$.
2. $f(A_1 \cap A_2) \subseteq f(A_1) \cap f(A_2)$, where equality holds if f is injective (one-to-one).

Proposition 11.3 (Lattice / Correspondence for Quotient Groups). *Let G be a group and $H \trianglelefteq G$. Then:*

1. *Any subgroup of G/H is of the form K/H , where K is a subgroup of G containing H ($H \leq K \leq G$).*
2. $K_1/H = K_2/H \implies K_1 = K_2$, where $K_1, K_2 \leq G$ containing H .
3. $(K_1 \cap K_2)/H = K_1/H \cap K_2/H$, where $K_1, K_2 \leq G$ containing H .

Proof. Consider the quotient group homomorphism $\nu : G \rightarrow G/H$ defined by $\nu(x) = xH$ for all $x \in G$. Then ν is surjective and $\ker \nu = H$.

By the general Correspondence Theorem (Theorem 1.1), there exists a bijection:

$$\phi : \mathcal{S}(G) \rightarrow \mathcal{S}'(G/H), \quad \phi(K) = \nu(K) \quad \forall K \in \mathcal{S}(G),$$

where $\mathcal{S}(G) = \{K \leq G \mid \ker \nu = H \subseteq K\}$ and $\mathcal{S}'(G/H) = \{T \leq G/H\}$.

Observe that for any $K \in \mathcal{S}(G)$:

$$\nu(K) = \{\nu(x) \mid x \in K\} = \{xH \mid x \in K\} = K/H.$$

1. Let $T \leq G/H$. Then $T \in \mathcal{S}'(G/H)$. Since ϕ is onto, there exists $K \in \mathcal{S}(G)$ such that:

$$T = \phi(K) = \nu(K) = K/H.$$

Since $K \in \mathcal{S}(G)$, we have $\ker \nu \subseteq K \implies H \leq K$. Thus $T = K/H$ for some $K \leq G$ containing H .

2. Let $K_1/H = K_2/H$.

$$\implies \nu(K_1) = \nu(K_2) \implies \phi(K_1) = \phi(K_2).$$

Since ϕ is injective (one-to-one):

$$K_1 = K_2.$$

3. Since ϕ is a bijection, we have:

$$\begin{aligned} (K_1 \cap K_2)/H &= \nu(K_1 \cap K_2) \\ &= \phi(K_1 \cap K_2) \\ &= \phi(K_1) \cap \phi(K_2) \quad (\text{as } \phi \text{ is a bijection}) \\ &= \nu(K_1) \cap \nu(K_2) \\ &= K_1/H \cap K_2/H. \end{aligned}$$

Therefore, $\boxed{(K_1 \cap K_2)/H = K_1/H \cap K_2/H}$. □

Note. If G is a group and $H \trianglelefteq G$, then any normal subgroup of G/H is of the form K/H , where K is a normal subgroup of G containing H ($H \subseteq K \trianglelefteq G$).

12 The Three Isomorphism Theorems

12.1 The First Isomorphism Theorem

Theorem 12.1 (Fundamental Theorem of Group Homomorphisms / First Isomorphism Theorem). *If G and G' are two groups and $f : G \rightarrow G'$ is a group homomorphism, then:*

$$G/\ker f \cong f(G).$$

In particular, if f is surjective (onto), then:

$$G/\ker f \cong G'.$$

Proof. As $f : G \rightarrow G'$ is a group homomorphism, $\ker f \trianglelefteq G$. Therefore, the quotient group $G/\ker f$ exists. Also, the image $f(G) = \text{Im } f$ is a subgroup of G' .

Define the map $\phi : G/\ker f \rightarrow f(G)$ by:

$$\phi(x \ker f) = f(x) \quad \forall x \in G.$$

We prove that ϕ is an isomorphism in four explicit steps:

Step 1: ϕ is well-defined. Let $x \ker f = y \ker f$ for some $x, y \in G$.

$$\begin{aligned} x \ker f = y \ker f &\implies x^{-1}y \in \ker f \\ &\implies f(x^{-1}y) = e' \\ &\implies f(x^{-1})f(y) = e' && \text{(as } f \text{ is a homomorphism)} \\ &\implies (f(x))^{-1}f(y) = e' \\ &\implies f(x) = f(y) \\ &\implies \phi(x \ker f) = \phi(y \ker f). \end{aligned}$$

Hence, ϕ is well-defined.

Step 2: ϕ is injective (one-to-one). Reversing the logic of Step 1:

$$\begin{aligned} \phi(x \ker f) = \phi(y \ker f) &\implies f(x) = f(y) \\ &\implies (f(x))^{-1}f(y) = e' \\ &\implies f(x^{-1})f(y) = e' \\ &\implies f(x^{-1}y) = e' \\ &\implies x^{-1}y \in \ker f \\ &\implies x \ker f = y \ker f. \end{aligned}$$

Therefore, ϕ is a one-to-one map.

Step 3: ϕ is surjective (onto). Let $t \in f(G)$. By definition of the image $f(G)$, there exists $a \in G$ such that:

$$t = f(a).$$

Since $a \in G \implies a \ker f \in G/\ker f$, and:

$$\phi(a \ker f) = f(a) = t.$$

So ϕ is onto.

Step 4: ϕ is a group homomorphism. For any $x \ker f, y \ker f \in G/\ker f$:

$$\begin{aligned}\phi(x \ker f \cdot y \ker f) &= \phi(xy \ker f) \\ &= f(xy) \\ &= f(x)f(y) \quad (\text{as } f \text{ is a group homomorphism}) \\ &= \phi(x \ker f)\phi(y \ker f).\end{aligned}$$

So ϕ is a group homomorphism.

Combining Steps 1–4, ϕ is an isomorphism from $G/\ker f$ onto $f(G)$. Therefore:

$$\therefore G/\ker f \cong f(G).$$

If f is onto, then $f(G) = G'$, and hence:

$$G/\ker f \cong G'.$$

□

12.2 Applications of the First Isomorphism Theorem

Example 12.1 (Application 1: $\mathbb{R}/\mathbb{Z} \cong S^1$). Let $(\mathbb{R}, +)$ be the additive group of real numbers, and let:

$$S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$$

be the circle group under complex multiplication. Show that $\mathbb{R}/\mathbb{Z} \cong S^1$.

Proof. Define $f : (\mathbb{R}, +) \rightarrow (S^1, \cdot)$ by:

$$f(x) = e^{2\pi i x} \quad \forall x \in \mathbb{R}.$$

1. **f is a group homomorphism:** For any $x, y \in \mathbb{R}$:

$$f(x + y) = e^{2\pi i(x+y)} = e^{2\pi i x} \cdot e^{2\pi i y} = f(x) \cdot f(y).$$

So f is a group homomorphism.

2. **f is onto:** Let $z \in S^1$. Since $|z| = 1$, z can be written in polar form as:

$$z = e^{i\theta} \quad \text{for some } \theta \in \mathbb{R}.$$

Setting $x = \frac{\theta}{2\pi} \in \mathbb{R}$, we find:

$$f\left(\frac{\theta}{2\pi}\right) = e^{2\pi i\left(\frac{\theta}{2\pi}\right)} = e^{i\theta} = z.$$

So f is onto.

3. **Computation of $\ker f$:**

$$\begin{aligned}\ker f &= \{x \in \mathbb{R} \mid f(x) = 1\} \\ &= \{x \in \mathbb{R} \mid e^{2\pi i x} = 1\} \\ &= \{x \in \mathbb{R} \mid \cos(2\pi x) + i \sin(2\pi x) = 1\} \\ &= \{x \in \mathbb{R} \mid 2\pi x = 2k\pi, k \in \mathbb{Z}\} \\ &= \{x \in \mathbb{R} \mid x \in \mathbb{Z}\} = \mathbb{Z}.\end{aligned}$$

By the Fundamental Theorem of Group Homomorphisms:

$$\mathbb{R}/\ker f \cong S^1 \implies \boxed{\mathbb{R}/\mathbb{Z} \cong S^1}. \quad \square$$

Example 12.2 (Application 2: $\mathbb{Q}/\mathbb{Z} \cong P$). Let $P = \{z \in \mathbb{C} \mid z^n = 1 \text{ for some } n \in \mathbb{Z}^+\}$ be the group of all roots of unity under complex multiplication. Show that $\mathbb{Q}/\mathbb{Z} \cong P$.

Proof. Define $f : (\mathbb{Q}, +) \rightarrow (P, \cdot)$ by:

$$f(x) = e^{2\pi i x} \quad \forall x \in \mathbb{Q}.$$

1. **f is well-defined into P :** If $x \in \mathbb{Q}$, then $x = \frac{m}{n}$ for some $m, n \in \mathbb{Z}$ with $n \neq 0$. Then:

$$(e^{2\pi i x})^n = (e^{2\pi i \frac{m}{n}})^n = e^{2\pi i m} = 1.$$

Thus $f(x) \in P$, so f is a well-defined map.

2. **f is a group homomorphism:** For all $x, y \in \mathbb{Q}$:

$$f(x+y) = e^{2\pi i(x+y)} = e^{2\pi i x} e^{2\pi i y} = f(x)f(y).$$

3. **f is onto:** Let $z \in P$. Then $z^n = 1$ for some $n \in \mathbb{Z}^+$. By Euler's formula:

$$\cos(2\pi m) + i \sin(2\pi m) = 1 \quad \forall m \in \mathbb{Z} \implies z^n = e^{2\pi i m} \implies z = e^{2\pi i \frac{m}{n}}.$$

Since $\frac{m}{n} \in \mathbb{Q}$, we have:

$$z = f\left(\frac{m}{n}\right) \quad \text{with } \frac{m}{n} \in \mathbb{Q}.$$

Therefore, f is onto.

4. **Computation of $\ker f$:**

$$\begin{aligned} \ker f &= \{x \in \mathbb{Q} \mid f(x) = 1\} \\ &= \{x \in \mathbb{Q} \mid e^{2\pi i x} = 1\} \\ &= \{x \in \mathbb{Q} \mid x \in \mathbb{Z}\} = \mathbb{Z}. \end{aligned}$$

By the Fundamental Theorem of Group Homomorphisms:

$$\mathbb{Q}/\ker f \cong P \implies \boxed{\mathbb{Q}/\mathbb{Z} \cong P}. \quad \square$$

Example 12.3 (Application 3: $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$). Show that $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$.

Proof. Define $f : (\mathbb{Z}, +) \rightarrow (\mathbb{Z}_n, +)$ by:

$$f(x) = \bar{x}.$$

1. **f is a group homomorphism:**

$$f(x_1 + x_2) = \overline{x_1 + x_2} = \bar{x}_1 + \bar{x}_2 = f(x_1) + f(x_2).$$

2. **f is clearly onto:** For every $\bar{r} \in \mathbb{Z}_n = \{\bar{0}, \bar{1}, \dots, \overline{n-1}\}$, $f(r) = \bar{r}$ with $r \in \mathbb{Z}$.

3. **Computation of $\ker f$:**

$$\begin{aligned}\ker f &= \{x \in \mathbb{Z} \mid f(x) = \bar{0}\} \\ &= \{x \in \mathbb{Z} \mid \bar{x} = \bar{0}\} \\ &= \{x \in \mathbb{Z} \mid x = nk, k \in \mathbb{Z}\} \\ &= n\mathbb{Z}.\end{aligned}$$

By the Fundamental Theorem of Group Homomorphisms:

$$\mathbb{Z}/\ker f \cong \mathbb{Z}_n \implies \boxed{\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n}.$$

□

Example 12.4 (Application 4: $GL(n, \mathbb{R})/SL(n, \mathbb{R}) \cong (\mathbb{R}^*, \cdot)$). Recall that:

$$\begin{aligned}GL(n, \mathbb{R}) &= \{A \in M_n(\mathbb{R}) \mid \det(A) \neq 0\}, \\ SL(n, \mathbb{R}) &= \{A \in GL(n, \mathbb{R}) \mid \det(A) = 1\}.\end{aligned}$$

Show that $GL(n, \mathbb{R})/SL(n, \mathbb{R}) \cong (\mathbb{R}^*, \cdot)$.

Proof. Define $f : GL(n, \mathbb{R}) \rightarrow (\mathbb{R}^*, \cdot)$ by:

$$f(A) = \det(A) \quad \forall A \in GL(n, \mathbb{R}).$$

1. **f is a group homomorphism:** For any $A, B \in GL(n, \mathbb{R})$:

$$f(AB) = \det(AB) = \det(A) \det(B) = f(A)f(B).$$

2. **f is onto:** For any $\alpha \in \mathbb{R}^*$, consider the diagonal matrix:

$$A = \begin{pmatrix} \alpha & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}_{n \times n} \in GL(n, \mathbb{R}).$$

Then $\det(A) = \alpha \cdot 1 \cdots 1 = \alpha$. Thus $f(A) = \alpha$, so f is onto.

3. **Computation of $\ker f$:**

$$\begin{aligned}\ker f &= \{A \in GL(n, \mathbb{R}) \mid f(A) = 1\} \\ &= \{A \in GL(n, \mathbb{R}) \mid \det(A) = 1\} \\ &= SL(n, \mathbb{R}).\end{aligned}$$

By the Fundamental Theorem of Group Homomorphisms:

$$GL(n, \mathbb{R})/\ker f \cong (\mathbb{R}^*, \cdot) \implies \boxed{GL(n, \mathbb{R})/SL(n, \mathbb{R}) \cong (\mathbb{R}^*, \cdot)}.$$

□

12.3 The Second Isomorphism Theorem (Diamond Isomorphism Theorem)

Theorem 12.2 (Second Isomorphism Theorem). *Let H and K be subgroups of a group G such that $K \trianglelefteq G$. Then:*

$$\frac{H}{H \cap K} \cong \frac{HK}{K}.$$

Proof. We construct the isomorphism by applying the First Isomorphism Theorem to an appropriately defined map on H :

1. As $K \trianglelefteq G$, by Theorem 1.2 we have $H \cap K \trianglelefteq H$. Therefore, the quotient group $\frac{H}{H \cap K}$ exists.
2. Since $K \trianglelefteq G$, HK is a subgroup of G and $K \leq HK$. Since $K \trianglelefteq G$, $K \trianglelefteq HK$. Therefore, the quotient group $\frac{HK}{K}$ exists.
3. Define $f : H \rightarrow HK/K$ by:

$$f(h) = hK \quad \forall h \in H.$$

Since $H \subseteq HK$, $hK \in HK/K$, so f is a well-defined map.

4. **f is a group homomorphism:** For any $h_1, h_2 \in H$:

$$\begin{aligned} f(h_1 h_2) &= (h_1 h_2)K \\ &= (h_1 K)(h_2 K) \\ &= f(h_1)f(h_2). \end{aligned}$$

Thus f is a group homomorphism.

5. **f is onto:** Let $t \in HK/K$. Then $t = xK$ for some $x \in HK$. Since $x \in HK$, $x = hk$ for some $h \in H$ and $k \in K$. Then:

$$t = xK = (hk)K = h(kK) = hK = f(h).$$

Hence, f is onto.

6. **Computation of $\ker f$:**

$$\begin{aligned} \ker f &= \{h \in H \mid f(h) = K\} \\ &= \{h \in H \mid hK = K\} \\ &= \{h \in H \mid h \in K\} \\ &= H \cap K. \end{aligned}$$

By the Fundamental Theorem of Group Homomorphisms:

$$H/\ker f \cong HK/K \implies \boxed{\frac{H}{H \cap K} \cong \frac{HK}{K}}.$$

□

12.4 The Third Isomorphism Theorem

Theorem 12.3 (Third Isomorphism Theorem). *Let H and K be normal subgroups of a group G such that $H \subseteq K$ ($H \trianglelefteq G, K \trianglelefteq G$, and $H \subseteq K$). Then:*

$$\frac{G/H}{K/H} \cong \frac{G}{K}.$$

Proof. 1. As $H, K \trianglelefteq G$, the quotient groups G/H and G/K exist. Also, $H \subseteq K$, so H is a normal subgroup of K . Therefore, the quotient group K/H exists. As $K \trianglelefteq G$ such that $H \subseteq K$, by the Correspondence Theorem $K/H \trianglelefteq G/H$. Therefore, the quotient group $\frac{G/H}{K/H}$ exists.

2. Define $f : G/H \rightarrow G/K$ by:

$$f(xH) = xK \quad \forall x \in G.$$

Well-definedness of f :

$$\begin{aligned} xH = yH &\implies x^{-1}y \in H \subseteq K \\ &\implies x^{-1}y \in K \\ &\implies xK = yK \\ &\implies f(xH) = f(yH). \end{aligned}$$

So f is a well-defined map.

3. **f is a group homomorphism:** For any $xH, yH \in G/H$:

$$\begin{aligned} f(xH \cdot yH) &= f(xyH) \\ &= xyK \\ &= xK \cdot yK \\ &= f(xH)f(yH). \end{aligned}$$

Thus f is a group homomorphism.

4. **f is onto:** Let $t \in G/K$. Then $t = xK$ for some $x \in G$. Since $x \in G \implies xH \in G/H$, and:

$$f(xH) = xK = t.$$

So f is onto.

5. **Computation of $\ker f$:**

$$\begin{aligned} \ker f &= \{xH \in G/H \mid f(xH) = K\} \\ &= \{xH \in G/H \mid xK = K\} \\ &= \{xH \in G/H \mid x \in K\} \\ &= K/H. \end{aligned}$$

By the Fundamental Theorem of Group Homomorphisms:

$$(G/H)/\ker f \cong G/K \implies \boxed{\frac{G/H}{K/H} \cong \frac{G}{K}}.$$

□

12.5 The Isomorphism $S_n/A_n \cong (\{1, -1\}, \cdot)$

Problem 12.5. Show that $S_n/A_n \cong (\{1, -1\}, \cdot)$.

Proof. Define the signature map $\text{sgn} : S_n \rightarrow (\{1, -1\}, \cdot)$ by:

$$\text{sgn}(\sigma) = \begin{cases} +1 & \text{if } \sigma \text{ is an even permutation,} \\ -1 & \text{if } \sigma \text{ is an odd permutation.} \end{cases}$$

1. Since the parity of a product of permutations satisfies $\text{sgn}(\sigma\tau) = \text{sgn}(\sigma)\text{sgn}(\tau)$, sgn is a group homomorphism.
2. Since $n \geq 2$, S_n contains transpositions (which are odd), so sgn is surjective onto $\{1, -1\}$.
3. The kernel is:

$$\ker(\text{sgn}) = \{\sigma \in S_n \mid \text{sgn}(\sigma) = 1\} = A_n.$$

By the First Isomorphism Theorem:

$$S_n / \ker(\text{sgn}) \cong \{1, -1\} \implies \boxed{S_n/A_n \cong (\{1, -1\}, \cdot)}.$$

□

13 Products of Subgroups and Order Formulas

13.1 Criterion for HK to be a Subgroup

Definition 13.1 (Product of Subgroups). Let H and K be two subgroups of a group G . The **product** of H and K is defined as the set:

$$HK = \{hk \mid h \in H, k \in K\}.$$

Theorem 13.1 (Subgroup Criterion for the Product HK). *Let H and K be subgroups of a group G . Then HK is a subgroup of G if and only if $HK = KH$:*

$$HK \leq G \iff HK = KH.$$

Proof. We prove both implications:

($\implies$) **Assume $HK \leq G$. We show $HK = KH$.**

1. Let $x \in KH$. Then $x = kh$ for some $k \in K$ and $h \in H$. Consider its inverse:

$$x^{-1} = (kh)^{-1} = h^{-1}k^{-1}.$$

Since $h \in H \implies h^{-1} \in H$ and $k \in K \implies k^{-1} \in K$, we have $x^{-1} \in HK$. Since HK is given to be a subgroup of G , the inverse of any element in HK also belongs to HK :

$$(x^{-1})^{-1} \in HK \implies x \in HK.$$

Therefore:

$$KH \subseteq HK.$$

2. Conversely, let $y \in HK$. Since $HK \leq G$, $y^{-1} \in HK$. Thus $y^{-1} = hk$ for some $h \in H, k \in K$. Taking the inverse:

$$y = (y^{-1})^{-1} = (hk)^{-1} = k^{-1}h^{-1}.$$

Since $k^{-1} \in K$ and $h^{-1} \in H$, $y \in KH$. Therefore:

$$HK \subseteq KH.$$

Combining both inclusions:

$$HK = KH.$$

($\impliedby$) **Assume $HK = KH$. We show $HK \leq G$.**

1. Since $e \in H$ and $e \in K$, $e = ee \in HK \neq \emptyset$.
2. **Closure under multiplication:** Let $x, y \in HK$. Then $x = h_1k_1$ and $y = h_2k_2$ for $h_1, h_2 \in H$ and $k_1, k_2 \in K$.

$$\begin{aligned} xy &= (h_1k_1)(h_2k_2) \\ &= h_1(k_1h_2)k_2. \end{aligned}$$

Since $k_1h_2 \in KH$ and $KH = HK$, there exist $h_3 \in H$ and $k_3 \in K$ such that $k_1h_2 = h_3k_3$. Substituting:

$$xy = h_1(h_3k_3)k_2 = (h_1h_3)(k_3k_2).$$

Since $H \leq G \implies h_1h_3 \in H$, and $K \leq G \implies k_3k_2 \in K$, it follows that:

$$xy \in HK.$$

3. Inverses: Let $x = hk \in HK$. Then:

$$x^{-1} = (hk)^{-1} = k^{-1}h^{-1} \in KH.$$

Since $KH = HK$, $x^{-1} \in HK$.

Therefore, $HK \leq G$. □

Theorem 13.2 (Product of a Subgroup and a Normal Subgroup). *Let $H, K \leq G$. If $K \trianglelefteq G$, then $HK = KH$, and consequently $HK \leq G$.*

Proof. Let $x \in HK \implies x = hk$ for some $h \in H, k \in K$. Notice that:

$$x = hk = (hkh^{-1})h.$$

Since $K \trianglelefteq G$, $hkh^{-1} \in K$. Let $k' = hkh^{-1} \in K$. Then:

$$x = k'h \in KH \implies HK \subseteq KH.$$

Similarly, let $y \in KH \implies y = kh$ for some $k \in K, h \in H$. Notice that:

$$y = kh = h(h^{-1}kh).$$

Since $K \trianglelefteq G$, $h^{-1}kh \in K$. Let $k'' = h^{-1}kh \in K$. Then:

$$y = hk'' \in HK \implies KH \subseteq HK.$$

Hence $HK = KH$. By the preceding theorem, $HK \leq G$. □

Note. If $H, K \leq G$ such that **either** $H \trianglelefteq G$ or $K \trianglelefteq G$, then $HK \leq G$ and $KH \leq G$.

13.2 Formula for the Order of HK

Theorem 13.3 (Order Formula for the Product of Finite Subgroups). *If H and K are finite subgroups of a group G , then:*

$$|HK| = \frac{|H||K|}{|H \cap K|}.$$

Proof. Notice that the product set HK can be written as the union of left cosets of K :

$$HK = \bigcup_{h \in H} hK.$$

We determine how many distinct cosets of K appear in this union. For any $h_1, h_2 \in H$:

$$\begin{aligned} h_1K = h_2K &\iff h_1^{-1}h_2 \in K \\ &\iff h_1^{-1}h_2 \in H \cap K && (\text{since } h_1, h_2 \in H \implies h_1^{-1}h_2 \in H) \\ &\iff h_1(H \cap K) = h_2(H \cap K). \end{aligned}$$

Therefore, the number of distinct left cosets of K of the form hK ($h \in H$) is equal to the number of distinct left cosets of $H \cap K$ in H . This number is precisely the index:

$$[H : H \cap K] = \frac{|H|}{|H \cap K|}.$$

Each left coset hK contains exactly $|K|$ elements, and distinct cosets are completely disjoint. Therefore, the total number of elements in HK is:

$$|HK| = [H : H \cap K] \cdot |K| = \frac{|H|}{|H \cap K|} \cdot |K| = \frac{|H||K|}{|H \cap K|}. \quad \square$$

13.3 Comprehensive Practice Problems on Subgroup Orders

Proposition 13.4 (Divisibility of Element Orders under Homomorphisms). *If $f : G \rightarrow G'$ is a group homomorphism and $a \in G$ is an element of finite order $|a| = n$, then $|f(a)|$ is finite and divides $|a|$:*

$$|f(a)| \mid |a|.$$

Furthermore, if f is injective (one-to-one), then $|f(a)| = |a|$.

Proof. Let $|a| = n \implies a^n = e$. Applying f :

$$\begin{aligned} f(a^n) &= f(e) \\ \implies \underbrace{f(a)f(a) \cdots f(a)}_{n \text{ times}} &= e' \\ \implies (f(a))^n &= e'. \end{aligned}$$

Since $(f(a))^n = e'$, the order $|f(a)|$ is finite and must divide n :

$$|f(a)| \mid n \implies |f(a)| \mid |a|.$$

If f is one-to-one: let $m = |f(a)|$. Then $(f(a))^m = e' \implies f(a^m) = f(e) \implies a^m = e \implies n \mid m$. Since $m \mid n$ and $n \mid m$, $m = n$, so $|f(a)| = |a|$. $\square$

Problem 13.2. Are $\mathbb{Z}_4$ and V_4 isomorphic?

Solution. **No.** $\mathbb{Z}_4$ is a cyclic group and contains an element of order 4 (namely $\bar{1}$ and $\bar{3}$, since $|\bar{1}| = 4$). However, V_4 contains no element of order 4 (every non-identity element in V_4 has order 2). Since an isomorphism must preserve element orders, $\mathbb{Z}_4 \not\cong V_4$. $\square$

Problem 13.3. Show that if $H \leq S_n$, then either all permutations in H are even, or exactly half of them are even.

Solution. Consider the intersection $H \cap A_n$. By Theorem 1.2, since $A_n \trianglelefteq S_n$, we have $H \cap A_n \trianglelefteq H$. By the Second Isomorphism Theorem:

$$\frac{H}{H \cap A_n} \cong \frac{HA_n}{A_n}.$$

Since $A_n \leq HA_n \leq S_n$ and $[S_n : A_n] = 2$, there are only two possibilities for the subgroup HA_n :

1. $HA_n = A_n \implies H \subseteq A_n$. In this case, every permutation in H is even.
2. $HA_n = S_n$. In this case:

$$[H : H \cap A_n] = [S_n : A_n] = 2.$$

Thus $|H| = 2|H \cap A_n|$, which means exactly half of the elements in H belong to A_n (i.e., are even). □

Problem 13.4. Let K be a proper subgroup of H , and H be a proper subgroup of G ($K \subsetneq H \subsetneq G$). If $|K| = 42$ and $|G| = 420$, what are the possible orders of H ?

Solution. By Lagrange's Theorem:

$$|K| \mid |H| \quad \text{and} \quad |H| \mid |G|.$$

Thus 42 divides $|H|$, and $|H|$ divides 420. Let $|H| = 42 \cdot m$. Then $42 \cdot m \mid 420 \implies m \mid \frac{420}{42} = 10$. The positive divisors of 10 are 1, 2, 5, 10. Since $K \subsetneq H$ is a *proper* subgroup, $|H| > |K| = 42 \implies m > 1$. Since $H \subsetneq G$ is a *proper* subgroup, $|H| < |G| = 420 \implies m < 10$. Thus $m \in \{2, 5\}$.

- If $m = 2$: $|H| = 42 \times 2 = 84$.
- If $m = 5$: $|H| = 42 \times 5 = 210$.

Therefore, the possible orders of H are 84 or 210. □

Problem 13.5. Let $H, K \leq G$. If $|H| = 12$ and $|K| = 32$, find the possible values of $|H \cap K|$.

Solution. Since $H \cap K \leq H$ and $H \cap K \leq K$, by Lagrange's Theorem:

$$|H \cap K| \mid |H| = 12 \quad \text{and} \quad |H \cap K| \mid |K| = 32.$$

Therefore, $|H \cap K|$ must divide the greatest common divisor of 12 and 32:

$$|H \cap K| \mid \gcd(12, 32) = 4.$$

The positive divisors of 4 are 1, 2, 4. Therefore, the possible values of $|H \cap K|$ are 1, 2, or 4. □

Problem 13.6. Let a and b be elements of a group G . If $|a| = 10$ and $|b| = 21$, show that $\langle a \rangle \cap \langle b \rangle = \{e\}$.

Solution. Let $x \in \langle a \rangle \cap \langle b \rangle$.

- $x \in \langle a \rangle \implies |x| \mid |a| = 10$.
- $x \in \langle b \rangle \implies |x| \mid |b| = 21$.

Thus:

$$|x| \mid \gcd(10, 21) = 1.$$

Since $|x|$ is a positive integer dividing 1, we have $|x| = 1$. The only element of order 1 in any group is the identity element e . Therefore, $x = e$, which proves:

$$\boxed{\langle a \rangle \cap \langle b \rangle = \{e\}}.$$

□

14 The Center of a Group and Factor Group Theorems

14.1 The $G/Z(G)$ Theorem

Theorem 14.1 (The $G/Z(G)$ Cyclic-to-Abelian Theorem). *Let G be a group and let $Z(G)$ denote its center:*

$$Z(G) = \{z \in G \mid zx = xz \quad \forall x \in G\}.$$

If $G/Z(G)$ is cyclic, then G is abelian (and hence $Z(G) = G$, so $|G/Z(G)| = 1$).

Proof. Assume that $G/Z(G)$ is cyclic. Then there exists some coset $aZ(G) \in G/Z(G)$ such that:

$$G/Z(G) = \langle aZ(G) \rangle.$$

Let $x, y \in G$. We must show that $xy = yx$.

Since $xZ(G)$ and $yZ(G)$ are elements of the cyclic group $G/Z(G) = \langle aZ(G) \rangle$, they must be powers of the generator $aZ(G)$. Thus, there exist integers $r, s \in \mathbb{Z}$ such that:

$$xZ(G) = (aZ(G))^r = a^r Z(G) \quad \text{and} \quad yZ(G) = (aZ(G))^s = a^s Z(G).$$

By definition of cosets, this implies:

$$x = a^r z_1 \quad \text{and} \quad y = a^s z_2 \quad \text{for some } z_1, z_2 \in Z(G).$$

Now compute the product xy :

$$\begin{aligned} xy &= (a^r z_1)(a^s z_2) \\ &= a^r (z_1 a^s) z_2 \\ &= a^r (a^s z_1) z_2 && (\text{as } z_1 \in Z(G) \implies z_1 a^s = a^s z_1) \\ &= a^{r+s} z_1 z_2. \end{aligned}$$

Next compute the product yx :

$$\begin{aligned} yx &= (a^s z_2)(a^r z_1) \\ &= a^s (z_2 a^r) z_1 \\ &= a^s (a^r z_2) z_1 && (\text{as } z_2 \in Z(G) \implies z_2 a^r = a^r z_2) \\ &= a^{s+r} z_2 z_1 \\ &= a^{r+s} z_1 z_2 && (\text{as } z_1, z_2 \in Z(G) \implies z_2 z_1 = z_1 z_2). \end{aligned}$$

Comparing both expressions:

$$xy = yx.$$

Since this holds for all $x, y \in G$, G is an abelian group. Consequently, every element commutes with all elements of G , so $Z(G) = G$. Thus $|G/Z(G)| = [G : Z(G)] = [G : G] = 1$. $\square$

14.2 Forbidden Prime Orders for $G/Z(G)$

Problem 14.1. Show that $|G/Z(G)| \neq p$, where p is any prime number.

Solution. Suppose, for contradiction, that $|G/Z(G)| = p$ for some prime p . Every group of prime order is cyclic. Therefore, $G/Z(G)$ is cyclic. By the $G/Z(G)$ theorem, if $G/Z(G)$ is cyclic, then G must be abelian. If G is abelian, then $Z(G) = G$, which implies:

$$|G/Z(G)| = \frac{|G|}{|Z(G)|} = \frac{|G|}{|G|} = 1.$$

This contradicts $|G/Z(G)| = p$, since p is prime ($p \geq 2 > 1$). Therefore, $|G/Z(G)| \neq p$. $\square$

14.3 Non-Abelian Groups of Order pq

Problem 14.2. Let G be a group of order $|G| = pq$, where p and q are prime numbers. If G is non-abelian, show that $|Z(G)| = 1$.

Solution. Since $Z(G)$ is a subgroup of G ($Z(G) \leq G$), by Lagrange's Theorem:

$$|Z(G)| \mid |G| = pq.$$

The positive divisors of pq are $1, p, q, pq$. Thus, the only possible values for $|Z(G)|$ are:

$$|Z(G)| \in \{1, p, q, pq\}.$$

We examine each case:

1. **Case 1:** $|Z(G)| = pq$. Then $|Z(G)| = |G| \implies Z(G) = G$, which means G is abelian. This contradicts the hypothesis that G is non-abelian. Thus $|Z(G)| \neq pq$.
2. **Case 2:** $|Z(G)| = p$. Then the index is:

$$|G/Z(G)| = \frac{|G|}{|Z(G)|} = \frac{pq}{p} = q.$$

Since q is prime, the quotient group $G/Z(G)$ has prime order, so $G/Z(G)$ is cyclic. By the $G/Z(G)$ theorem, G must be abelian, which is a contradiction. Thus $|Z(G)| \neq p$.

3. **Case 3:** $|Z(G)| = q$. Then the index is:

$$|G/Z(G)| = \frac{|G|}{|Z(G)|} = \frac{pq}{q} = p.$$

Since p is prime, $G/Z(G)$ is cyclic, which again implies G is abelian (a contradiction). Thus $|Z(G)| \neq q$.

Eliminating p, q , and pq , the only remaining possibility is:

$$|Z(G)| = 1.$$

$\square$

14.4 Combinatorics of Cycles and Normal Subgroups of S_n

Theorem 14.2 (Formula for the Number of r -Cycles in S_n). *The number of distinct cycles of length r in the symmetric group S_n ($2 \leq r \leq n$) is:*

$$\binom{n}{r}(r-1)! = \frac{n!}{r!(n-r)!}(r-1)! = \frac{n!}{r(n-r)!} = \frac{{}^nP_r}{r}.$$

Proof. To construct an r -cycle $(a_1 a_2 \dots a_r)$:

1. Choose r distinct elements from $\{1, 2, \dots, n\}$ in $\binom{n}{r}$ ways.
2. Arrange these r elements in a circular permutation in $(r-1)!$ ways.

Multiplying yields $\binom{n}{r}(r-1)! = \frac{{}^nP_r}{r}$. □

Note (Normal Subgroups of S_n). • For all $n \neq 4$ ($n \geq 3$): the **only** normal subgroups of S_n are the trivial subgroup $\{I\}$, the alternating group A_n , and S_n itself:

$$\{I\}, \quad A_n, \quad S_n.$$

- For $n = 4$: the normal subgroups of S_4 are:

$$\{I\}, \quad V_4, \quad A_4, \quad S_4.$$

15 Commutator Subgroups (Derived Subgroups)

15.1 Definitions

Definition 15.1 (Commutator). Let G be a group and let $a, b \in G$. The element:

$$[a, b] = aba^{-1}b^{-1}$$

is called the **commutator** of the pair a and b .

Note. Observe that:

$$aba^{-1}b^{-1} = e \iff ab = ba.$$

Thus, commutators quantify the extent to which two elements fail to commute.

Definition 15.2 (Commutator / Derived Subgroup). Let S be the set of all commutators of elements of G :

$$S = \{aba^{-1}b^{-1} \mid a, b \in G\}.$$

The subgroup generated by S , denoted by $\langle S \rangle$, is called the **commutator subgroup** (or **derived subgroup**) of G . It is denoted by:

$$G' \quad \text{or} \quad G^c \quad \text{or} \quad [G, G].$$

15.2 The Five Fundamental Theorems on Commutators

Theorem 15.1 (The Five Fundamental Theorems on Commutators). *Let G be a group and G' its commutator subgroup. Then:*

1. $G' = \{e\}$ if and only if G is an abelian group.
2. $G' \trianglelefteq G$.
3. G/G' is an abelian group (called the **abelianization** of G).
4. If $H \trianglelefteq G$, then G/H is abelian if and only if $G' \subseteq H$.
5. Let $H \leq G$. If $G' \subseteq H$, then $H \trianglelefteq G$.

Proof. We prove each of the five statements rigorously:

Proof of (1): $G' = \{e\} \iff G$ is abelian.

- ($\implies$) Let $G' = \{e\}$. For any $a, b \in G$, the commutator $aba^{-1}b^{-1} \in S \subseteq G' = \{e\}$.

$$\begin{aligned} aba^{-1}b^{-1} = e &\implies aba^{-1} = b \\ &\implies ab = ba. \end{aligned}$$

Thus, G is abelian.

- ($\impliedby$) Let G be abelian. Then for all $a, b \in G$:

$$ab = ba \implies aba^{-1}b^{-1} = e.$$

Thus $S = \{e\}$, so $G' = \langle S \rangle = \langle \{e\} \rangle = \{e\}$.

Proof of (2): $G' \trianglelefteq G$. Let $g \in G$ and $h \in G'$. Since every element of G' is a finite product of commutators and their inverses, it suffices to prove that the conjugate of any commutator $aba^{-1}b^{-1} \in S$ by g is again in S :

$$\begin{aligned} g(aba^{-1}b^{-1})g^{-1} &= (gag^{-1})(gbg^{-1})(ga^{-1}g^{-1})(gb^{-1}g^{-1}) \\ &= (gag^{-1})(gbg^{-1})(gag^{-1})^{-1}(gbg^{-1})^{-1}. \end{aligned}$$

Setting $x = gag^{-1} \in G$ and $y = gbg^{-1} \in G$, we have:

$$g(aba^{-1}b^{-1})g^{-1} = xyx^{-1}y^{-1} \in S \subseteq G'.$$

Therefore, $gG'g^{-1} \subseteq G'$ for all $g \in G$, which proves $G' \trianglelefteq G$.

Proof of (3): G/G' is an abelian group. Let $aG', bG' \in G/G'$. We must show that $(aG')(bG') = (bG')(aG')$. Notice that:

$$ab = ba(a^{-1}b^{-1}ab) = ba[a^{-1}, b^{-1}].$$

Since $[a^{-1}, b^{-1}] \in S \subseteq G'$, we have:

$$(a^{-1}b^{-1}ab)G' = G'.$$

Therefore:

$$\begin{aligned} (aG')(bG') &= abG' \\ &= ba(a^{-1}b^{-1}ab)G' \\ &= baG' \\ &= (bG')(aG'). \end{aligned}$$

Hence, G/G' is an abelian group.

Proof of (4): If $H \trianglelefteq G$, then G/H is abelian $\iff G' \subseteq H$.

- ($\implies$) Let G/H be abelian. For any $a, b \in G$:

$$\begin{aligned} (aH)(bH) &= (bH)(aH) \implies abH = baH \\ &\implies (ba)^{-1}(ab) \in H \\ &\implies a^{-1}b^{-1}ab \in H \\ &\implies [a^{-1}, b^{-1}] \in H. \end{aligned}$$

Since this holds for all elements $a, b \in G$, every commutator belongs to H . Thus $S \subseteq H$. Since H is a subgroup containing S , it must contain the subgroup generated by S :

$$\langle S \rangle = G' \subseteq H.$$

- ($\impliedby$) Conversely, let $G' \subseteq H$. For any $a, b \in G$, the commutator $aba^{-1}b^{-1} \in G' \subseteq H$.

$$\begin{aligned} aba^{-1}b^{-1} \in H &\implies ab(ba)^{-1} \in H \\ &\implies abH = baH \\ &\implies (aH)(bH) = (bH)(aH). \end{aligned}$$

Thus, G/H is abelian.

Proof of (5): Let $H \leq G$. If $G' \subseteq H$, then $H \trianglelefteq G$. Let $a \in G$ and $h \in H$. Consider the commutator of a and h :

$$aha^{-1}h^{-1} \in G'.$$

Since $G' \subseteq H$, we have $aha^{-1}h^{-1} \in H$. Now multiply by $h \in H$ on the right:

$$aha^{-1} = (aha^{-1}h^{-1})h \in H \cdot H = H.$$

Thus $aha^{-1} \in H$ for all $a \in G$ and all $h \in H$. Therefore:

$$aHa^{-1} \subseteq H \implies H \trianglelefteq G. \quad \square$$

Remark (Minimality of G' Among Normal Subgroups with Abelian Quotient). From properties (3) and (4), we deduce that:

- G' is the **smallest** normal subgroup of G such that G/G' is abelian.
- G/G' is the **largest** abelian quotient group of G .

15.3 Explicit Computations of Commutator Subgroups

Example 15.3 (Commutator Subgroup of the Quaternion Group Q_8). Find the commutator subgroup of $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$.

Solution. Recall that $Z(Q_8) = \{1, -1\}$. The quotient group has order:

$$[Q_8 : \{1, -1\}] = \frac{8}{2} = 4.$$

Every group of order 4 is abelian (isomorphic to either $\mathbb{Z}_4$ or V_4). Since $Q_8/\{1, -1\}$ is abelian, by Property (4) of Theorem 7.1:

$$Q'_8 \subseteq \{1, -1\}.$$

Since Q_8 is non-abelian ($ij = k \neq -k = ji$), by Property (1):

$$Q'_8 \neq \{1\}.$$

Since the only subsets of $\{1, -1\}$ are $\{1\}$ and $\{1, -1\}$, we must have:

$$\boxed{Q'_8 = \{1, -1\} = Z(Q_8)}. \quad \square$$

Example 15.4 (Commutator Subgroup of S_3). Find the commutator subgroup of the symmetric group S_3 .

Solution. $S_3 = \{I, (12), (13), (23), (123), (132)\}$. The normal subgroups of S_3 are $\{I\}, A_3, S_3$.

1. Since S_3 is non-abelian, $S'_3 \neq \{I\}$.
2. The alternating group A_3 has index 2:

$$|S_3/A_3| = \frac{6}{3} = 2.$$

Every group of order 2 is cyclic, hence abelian. Thus S_3/A_3 is abelian.

3. By Property (4):

$$S'_3 \subseteq A_3.$$

4. The order of A_3 is 3 (prime), so its only subgroups are $\{I\}$ and A_3 . Since $S'_3 \neq \{I\}$, we conclude:

$$\boxed{S'_3 = A_3}. \quad \square$$

Example 15.5 (Commutator Subgroup of A_3). Find the commutator subgroup of A_3 .

Solution. $|A_3| = 3$, which is a prime number. Every group of prime order is cyclic, and every cyclic group is abelian. Since A_3 is abelian, by Property (1):

$$\boxed{A'_3 = \{I\}}. \quad \square$$

Example 15.6 (Commutator Subgroup of S_4). Find the commutator subgroup of S_4 .

Solution. The normal subgroups of S_4 are $\{I\}, V_4, A_4, S_4$.

1. Since $|S_4/A_4| = \frac{24}{12} = 2$, S_4/A_4 is abelian. By Property (4):

$$S'_4 \subseteq A_4.$$

2. Now consider the commutator of $(1\ 2)$ and $(2\ 3)$ in S_4 :

$$\begin{aligned} [(1\ 2), (2\ 3)] &= (1\ 2)(2\ 3)(1\ 2)^{-1}(2\ 3)^{-1} \\ &= (1\ 2)(2\ 3)(1\ 2)(2\ 3) \\ &= (1\ 2\ 3)(1\ 2\ 3) \\ &= (1\ 3\ 2). \end{aligned}$$

Thus the 3-cycle $(1\ 3\ 2) \in S'_4$.

3. Since $S'_4 \leq S_4$ and contains a 3-cycle, by the Conjugacy Theorem in Section 1, S'_4 must contain all 3-cycles in S_4 .

4. The 3-cycles generate the entire alternating group A_4 . Therefore:

$$A_4 \subseteq S'_4.$$

Combining $S'_4 \subseteq A_4$ and $A_4 \subseteq S'_4$, we obtain:

$$\boxed{S'_4 = A_4}. \quad \square$$

Example 15.7 (Commutator Subgroup of A_4). Find the commutator subgroup of A_4 .

Solution. The normal subgroups of A_4 are $\{I\}, V_4, A_4$.

1. Since A_4 is non-abelian, $A'_4 \neq \{I\}$.

2. Consider the quotient by the Klein four-group V_4 :

$$|A_4/V_4| = \frac{12}{4} = 3.$$

Since 3 is prime, A_4/V_4 is cyclic, hence abelian.

3. By Property (4):

$$A'_4 \subseteq V_4.$$

4. Since $A'_4 \trianglelefteq A_4$ and $A'_4 \subseteq V_4$, A'_4 is a normal subgroup of A_4 contained in V_4 . The only normal subgroups of A_4 are $\{I\}$, V_4 , and A_4 . Since $A'_4 \neq \{I\}$ and $A'_4 \subseteq V_4$, we must have:

$$\boxed{A'_4 = V_4}.$$

□

16 Introduction to Group Actions

16.1 Definition and Axioms of a Group Action

Definition 16.1 (Group Action). Let G be a group and let A be a non-empty set. A **group action** (or **action**) of G on A is a mapping:

$$* : G \times A \rightarrow A, \quad (g, a) \mapsto g * a$$

such that the following two axioms are satisfied:

1. **Compatibility (Associativity):**

$$g_1 * (g_2 * a) = (g_1 g_2) * a \quad \forall g_1, g_2 \in G, \forall a \in A.$$

2. **Identity:**

$$e * a = a \quad \forall a \in A,$$

where e denotes the identity element of G .

Definition 16.2 (G -Set). When an action of a group G on a set A is defined, the set A is called a G -set, and we say that G **acts on** A .

16.2 Left Actions vs. Right Actions

Definition 16.3 (Left Group Action). The action defined above, where group elements operate on elements of A from the left ($* : G \times A \rightarrow A$), is referred to as an **action on A by left** or a **left action** of G on A .

Definition 16.4 (Right Group Action). We can also define an action of G on A from the right by considering a mapping:

$$* : A \times G \rightarrow A, \quad (a, g) \mapsto a * g$$

satisfying the corresponding right action axioms:

1. **Compatibility:**

$$(a * g_1) * g_2 = a * (g_1 g_2) \quad \forall g_1, g_2 \in G, \forall a \in A.$$

2. **Identity:**

$$a * e = a \quad \forall a \in A.$$